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Practical Applications of Water Hammer Analysis from Hydraulic Fracturing Treatments

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Abstract

Water hammer is oscillatory pressure behavior in a wellbore resulting from the inertial effect of flowing fluid being subjected to an abrupt change in velocity. It is commonly observed at the end of large-scale hydraulic fracturing treatments after fluid injection rate is rapidly reduced or terminated.

In this paper, factors affecting treatment-related water hammer behavior are disclosed, and field studies are introduced correlating water hammer characteristics to fracture intensity and well productivity.

A simulator based on fundamental fluid-mechanics concepts was developed to model water hammer responses for various wellbore configurations and treatment characteristics. Insight from the modeling work was used to develop an optimal process of terminating fluid injection to obtain a consistent, identifiable oscillatory response for evaluating water hammer periodicity, decay rate, and oscillatory patterns.

A completion database was engaged in a semi-automated process to evaluate numerous treatments. A data screening method was developed and implemented for enhancing interpretation reliability. Derived water hammer components were correlated to fracture intensity, well productivity and in certain cases, loss of treatment confinement to the intended treatment interval. Using the above process, thousands of hydraulic fracturing treatments were evaluated, and the results of that work are included in this study. The treatments were performed in wells based in Texas, South America, and Canada and completed in low permeability and unconventional reservoirs.

The water hammer decay rate was determined to be a reliable indication of the system friction (friction in the wellbore and hydraulic fracture network) that drains energy from the water hammer pulse. In unconventional reservoirs characterized by small differences in the minimum and maximum horizontal stresses, high system friction correlated positively with fracture intensity/complexity and well performance. Results were constrained with instantaneous shut-in pressure (ISIP) and pressure falloff measurements to identify instances of direct communication with previously treated offset wellbores. The resulting analyses provided:

- identification of enhanced-permeability intervals
- indications of hydraulic fracture geometry
- assessment of treatment modifications intended to enhance fracture complexity
- identification of loss of treatment confinement to the intended interval

- location of associated points of failure in the wellbore

Topics covered in the paper include:

- Introduction
 - Joukowsky Equation
 - Period and Boundary Conditions
- Review of Prior Work on Water Hammer Analysis
- Shut-In Pressure Data, Analysis, and Model
 - Data collection frequency
 - Data issues and requirements
 - Water Hammer Analytical Method
 - Water Hammer Model
- Effects on Water hammer signature
 - Fluid properties
 - Step-down rate change and duration
 - Perforation friction
- Applications
 - Identification of Boundary Condition
 - Identification of Treatment Stage Isolation
 - Identification of Casing Failure Depth
 - Identification of Excess Period (Excess Length)
- Case Study – Water Hammer Data in an Unconventional Reservoir
 - Interpretation of frac geometry and friction in the fracture
 - Relationship to well productivity

Introduction

Water hammer occurs when there is a fast change in operating conditions for a well or pipeline. This may involve the sudden closing of a valve or change in injection or production rate. In this paper, the focus is for rate step-downs or termination (shut-in) conducted near the end of fracturing treatments.

For routine hydraulic fracturing applications, the following are the steps that result in the water hammer signature (see [Fig. 1](#)):

- Pump trucks inject fracturing fluid and proppant into the well.
- Upon sudden rate reduction and/or pump shutoff, a pressure pulse is measured at the wellhead. For [Fig. 1](#), two rate reductions were conducted. The first was to half rate; the second was to zero rate. With each rate reduction, separate water hammer signatures resulted.

- This pulse moves from the surface down through the wellbore, interacts with the created hydraulic fractures, and is reflected up the wellbore. This process will repeat periodically until energy is drained from the pulse.

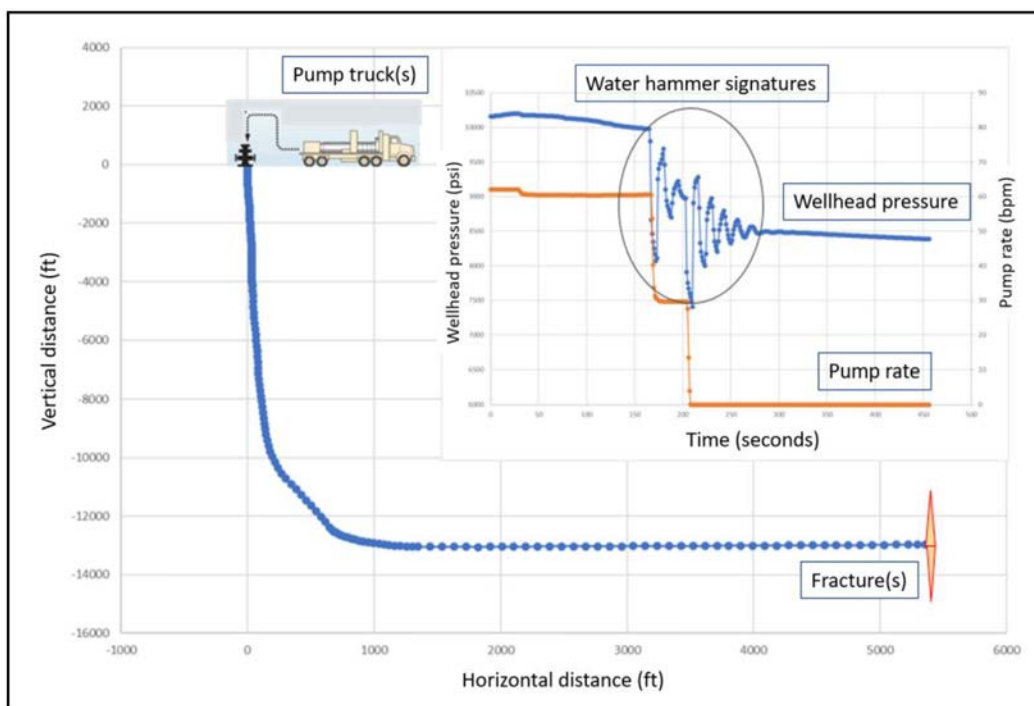


Figure 1—Schematic of a well, hydraulic fracture treatment, and water hammer signature.

The water hammer pressure signature is the result of the conversion of the kinetic energy of the fluid to potential energy when the surface injection rate is sharply reduced or terminated. The potential energy change is expressed as a sudden increase or decrease of fluid pressure.

Joukowsky Equation

Fig. 2 shows a pipe carrying fluid moving at a speed ΔV with a density of ρ and pressure of P which is stopped by a fast-closing valve from a fixed frame of reference. This sudden closure leads to:

- a velocity decrease to 0,
- a density increase of $\rho + \Delta\rho$,
- a pressure increase of $P + \Delta P$ upstream of the valve,
- and the creation of a pressure wave (indicated by the dashed line) moving from right to left at the fluid speed of sound, C .

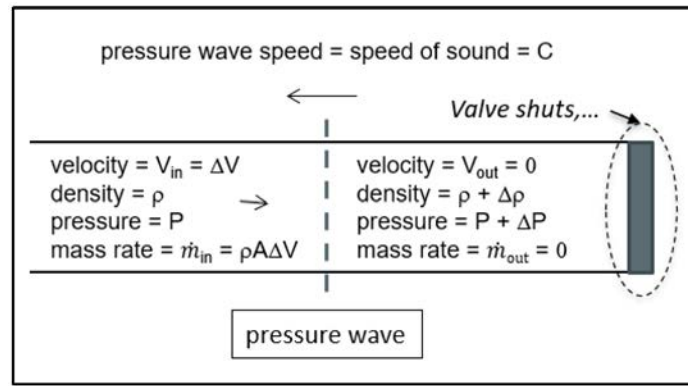


Figure 2—Pipe carrying fluid with a fast closing valve (fixed frame).

Fig. 3 shows the same concept as Fig. 2, but the difference is the frame of reference. Fig. 2 is a fixed frame of reference while Fig. 3 is a moving frame of reference where the coordinate system moves with the pressure wave at the speed of sound. The pressure wave is indicated by the dashed vertical line. For the modeling concept of using a moving frame reference, the mass rate is the same upstream and downstream of the pressure wave.

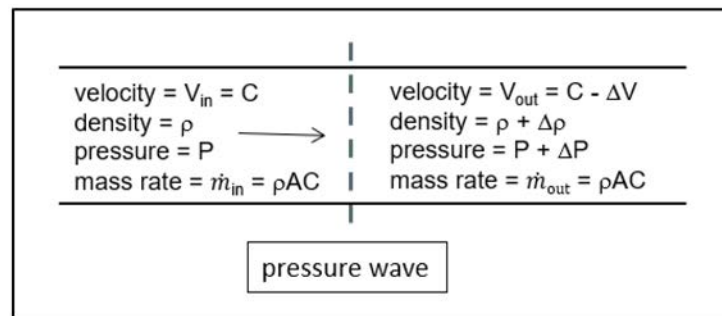


Figure 3—Pipe carrying fluid with a fast closing valve (moving frame).

Applying a force balance across the pressure wave in the moving frame,

$$F = \dot{m}(V_{out} - V_{in}) = (\rho AC)(C - \Delta V - C) = -\rho C \Delta V \quad (1)$$

Finally,

$$\Delta P = F/A = -\rho C \Delta V \quad (2)$$

The equation above is the Joukowsky equation, which relates the pressure change ΔP in response to a change in velocity ΔV . The pressure change ΔP can be either positive or negative, depending on how it was created. For example, for a sudden valve closure in the middle of a wellbore where fluid was being pumped down the wellbore, there will be a pressure increase upstream of the valve as pressure 'piles up' against the closed valve. There will also be a corresponding pressure decrease downstream of the valve as fluid moving downstream of the closed valve 'pulls' on the fluid that has been stopped by the closed valve.

The resultant pressure wave created by the water hammer event moves at the speed of sound of the fluid through the wellbore (adjusted to accommodate wellbore and multiphase effects as necessary). This pressure wave then reflects off wellbore diameter reductions, leaks, perforations, and ultimately the hydraulic fracture system.

Period and Boundary Conditions

Depending on the nature of the boundary condition imposed at the bottom of the well, the periodicity of the water hammer signature at the top of the well induced by the injection pump step-down will change significantly. For unconventional reservoirs characterized by low and ultra-low permeability, the following are examples of the two boundary condition scenarios. The following inputs were used in the company's developed water hammer simulator.

- Well length is 6,000 m (19,694 ft)
- Well diameter is 11.86 cm (4.67 inch)
- The fluid density is 1,000 kg/m³ (8.34 lb/gal)
- Fluid speed of sound: 1,500 m/s (4,920 ft/s)
- For simplification, hydrostatic pressure variations within the wellbore are not considered.

Scenario 1 – Closed inlet and constant pressure outlet. This condition exists during shut-in at the end of a treatment, where hydraulic fractures were created thereby having large fracture capacity (closed inlet = shut-in of well at the surface; constant pressure outlet = large fracture capacity at the bottom of the wellbore). Fig. 4 shows the behavior of a well which is closed at the inlet while maintaining a constant pressure at the outlet. After 1.5 seconds into the shut-in, a pressure deficit is created at the inlet of the well. The fluid has stopped near the inlet (velocity equals zero) but is moving elsewhere further down from the inlet.

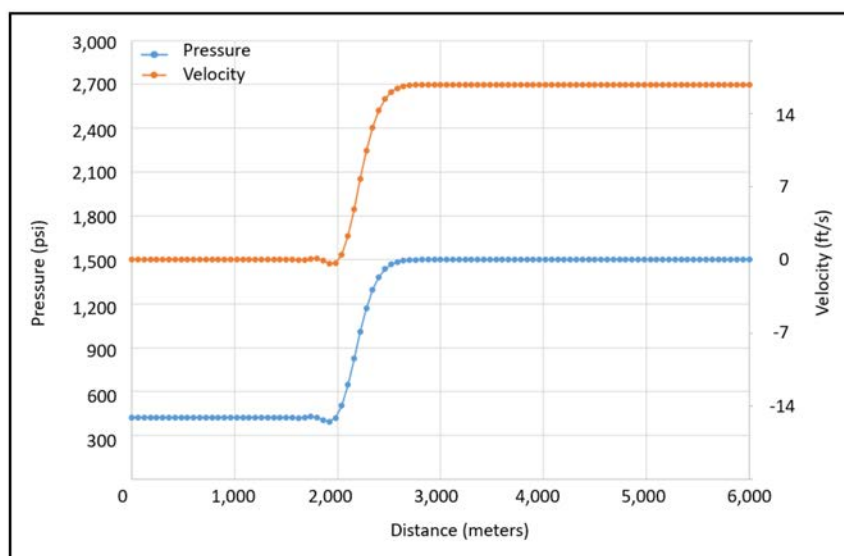


Figure 4—Pressure and velocity vs. wellbore length from inlet, 1.5 seconds into the shut-in: closed inlet, constant pressure outlet.

As indicated in Fig. 5, the wave pattern repeats itself every 16 seconds, meaning that a pressure wave moving at 1,500 m/s will make two round trips back and forth through the wellbore per cycle (per period).

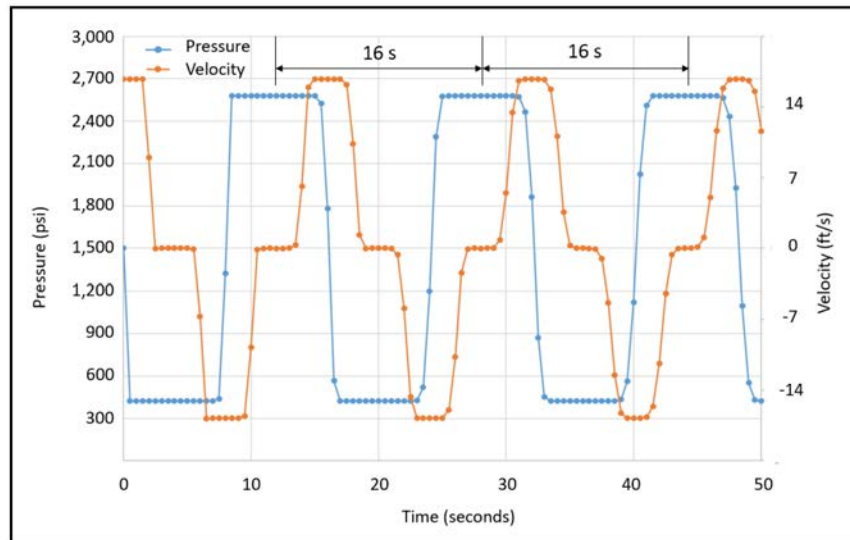


Figure 5—Wellhead (inlet) pressure and velocity as a function of time: closed inlet, constant pressure outlet.

Fig. 6 provides a visual to further explain the relationship between this boundary condition (closed inlet and constant pressure outlet) and the water hammer period.

- The time for the pressure wave to travel the distance of the 6,000 m pipeline would be the length of the wellbore divided by the fluid speed of sound ($6,000 \text{ m} / (1,500 \text{ m/s}) = 4 \text{ seconds}$).
- Two round trips equal four times the length of the pipeline.
- The cycle or period is equal to four times the wave travel time for the length of the pipeline. (For this example, $4 \times 4 \text{ s} = 16 \text{ seconds}$.)

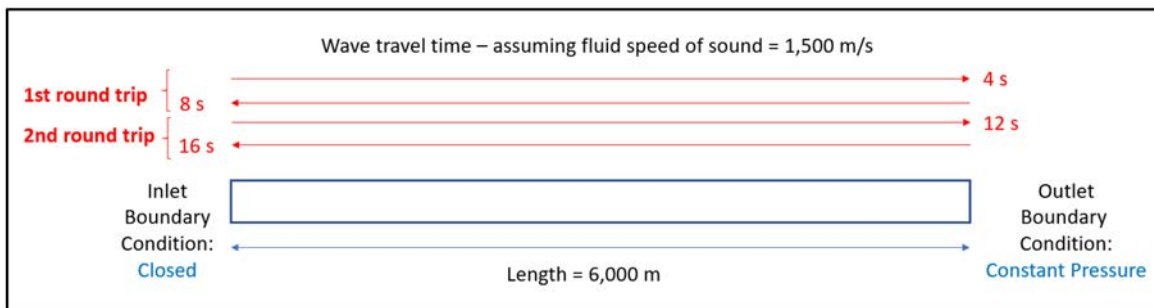


Figure 6—Schematic of wave travel time for one water hammer cycle (period): closed inlet, constant pressure outlet.

Scenario 2 – Closed inlet and closed outlet. Field examples of this condition include shut-in as the result of a screen out event (closed inlet = shut-in of well at the surface; closed outlet = screen out at the bottom of the wellbore) and generated shock waves (e.g., perforating event) when there is nil fracture capacity at the wellbore outlet.

Fig. 7 shows the behavior of a well that is closed at the inlet and the outlet. After 1.5 seconds into the shut-in, a pressure deficit is created at the inlet of the well. The fluid has stopped near the inlet and the outlet (velocity equals zero) but is still moving forward in the middle of the pipe.

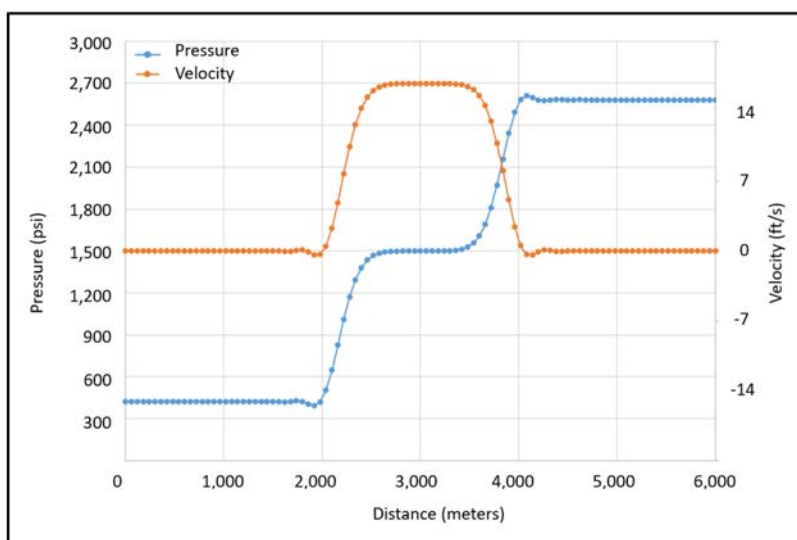


Figure 7—Pressure and velocity vs. wellbore length from inlet, 1.5 seconds into the shut-in: closed inlet, closed outlet.

As indicated in Fig. 8, the wave pattern repeats itself every 8 seconds, meaning that a pressure wave moving at 1,500 m/s will make one round trip back and forth through the pipeline per cycle (per period).

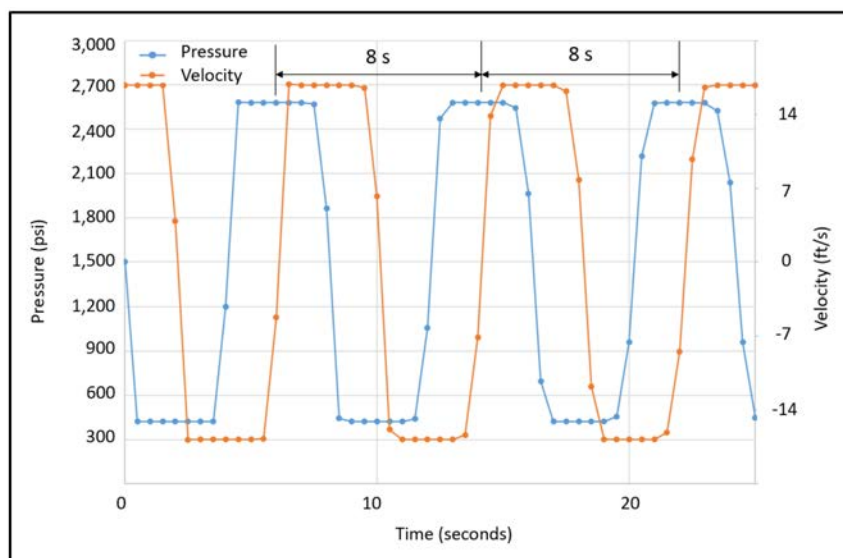


Figure 8—Wellhead (inlet) pressure and velocity as a function of time: closed inlet, closed outlet.

Fig. 9 provides a visual to further explain the relationship between this boundary condition (closed inlet and closed outlet) and the water hammer period.

- The time for the pressure wave to travel the distance of the 6,000 m pipeline would be the length of the wellbore divided by the fluid speed of sound ($6,000 \text{ m} / (1,500 \text{ m/s}) = 4 \text{ seconds}$).
- One round trip equals two times the length of the pipeline.
- For the boundary conditions of closed inlet and closed outlet, the pressure period is equal to two times the wave travel time for the length of the pipeline. (For this example, $2 \times 4 \text{ s} = 8 \text{ seconds}$.)

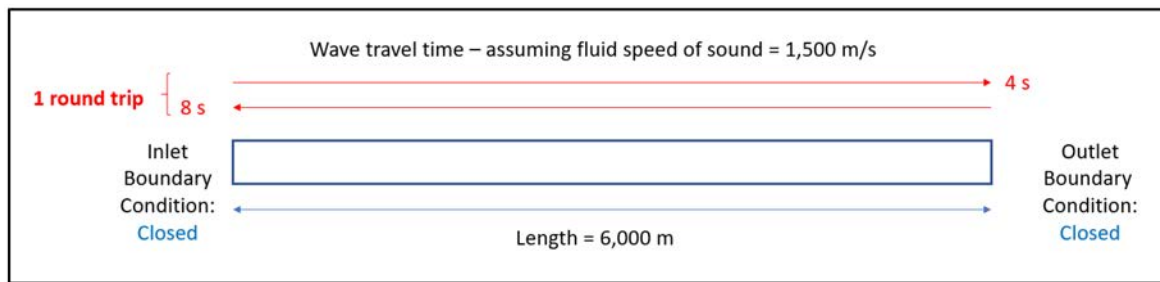


Figure 9—Schematic of wave travel time for one water hammer cycle (period): closed inlet, closed outlet.

Water Hammer Period Calculation. The following equation can be used to calculate the expected period for a water hammer signature.

$$\text{Period (sec)} = B \times MD / C \quad (3)$$

Where B = Boundary Condition factor

B = 4 for closed inlet and constant pressure outlet

B = 2 for closed inlet and closed outlet

MD = measured depth to flow exit (such as the perforation depth), ft or m

C = fluid speed of sound in the wellbore, ft/s or m/s.

Review of Prior Work on Water Hammer Analysis

The use of water hammer signatures as a cost-effective, scalable diagnostic solution to characterize aspects of hydraulically induced fractures has been of great interest to the industry and academic communities. The properties of the signal can indicate the quality of the connection between the wellbore, the fracture network, and the reservoir.

Holzhausen and Gooch (1985) first introduced the idea of using water-hammer signatures for fracture diagnostics, under the term impedance analysis. The method, referred to in later publications as Hydraulic Impedance Testing or HIT, relies on a lumped resistance-capacitance model to evaluate hydraulic fracture dimensions from changes in downhole impedance at the well-fracture interface. The model is analogous to an electrical circuit, where resistance (R) and capacitance (C) elements are combined in series, and fracture impedance is expressed as a function of flow resistance and fluid storage. An additional inertance term (I), describing the pressure difference required to cause a unit change in the rate of change of volumetric flow rate with time, was later added to the model formulation (Paige, 1992).

The technique was evaluated experimentally by Paige et al. (1995) and performed in water injection wells and mini-fracs (Holzhausen and Egan, 1986), where the interpreted fracture dimensions were compared to traditional well tests and reservoir simulations. Fracture length is calculated assuming the pulse transmitted into the fracture is reflected at the tip and by estimating excess travel time beyond the perforations. Wave speed is significantly lower in the fracture compared to the wellbore because of increased compliance, impacting travel time in the fracture. Fracture dimensions (width, height, and length) are interrelated through fracture compliance, which can be expressed analytically (Sneddon, 1946) for a semi-infinite fracture ($L_f \gg h_f$).

While early efforts were directed primarily toward fractured vertical wells, recent studies assessed the applicability of the HIT methodology to characterize hydraulic fractures in modern horizontal well completions. Mondal (2010) modeled the presence of multiple hydraulic fractures connected to the wellbore in any given fracturing stage by multiple capacitance elements in parallel, and solved water-hammer equations numerically using the explicit method of characteristics (MOC). By lumping the effect of multiple fractures into a single equivalent fracture, Carey et al. (2015) was able to characterize the average dimensions of the individual fractures in various field examples. Carey et al. (2016) also highlighted the

impact of R , C , I values on the simulated water-hammer signatures, and correlated them with microseismic surveys and production logs. [Hwang et al. \(2017\)](#) further extended the method to multi-stage hydraulic fracture treatments by accounting for mechanical stress interference in successive treatment stages.

[Ma et al. \(2019\)](#) proposed a new analytical formulation of water hammer pressure oscillation including pressure-dependent leak-off and perforation friction to determine fracture growth and near wellbore tortuosity. The boundary condition was derived through a fracture entry friction equation instead of using an electrical-circuit analogous system.

Another approach consists of recording reflected low-frequency tube waves generated at the wellhead and analyzing their interaction with fractures intersecting a wellbore in the frequency domain ([Dunham et al. 2017](#); [Liang et al. 2017](#)). By quantifying amplitude ratios and tube-wave attenuation over a range of frequencies, [Bakku et al. \(2013\)](#) were able to estimate the compliance, aperture, and lateral extent of a fluid-filled fracture intersecting a wellbore. [Dunham et al. \(2017\)](#) applied the concept of fracture impedance to estimate created hydraulic fracture conductivity. Following a similar methodology, [Clark et al. \(2018\)](#) focused on the frequency characteristics of hydraulic impulse events.

While many of the proposed models have been successful in recreating and matching water hammer signatures, it appears the optimization problem is ill-constrained, leading to non-unique solutions. The number of physical relationships is insufficient to resolve the variables of interest, such as fracture length, height, and width. The range of fracture geometry predictions for a particular stage is often shown to be broad despite matching the water hammer waveform. While the analysis of water hammer signatures is unlikely by itself to resolve the fracture geometry, combining it with various other analyses of pressure signatures in treatment well data (e.g., ISIP, net pressure) could provide additional constraints and help narrow down the range of solutions.

Shut-In Pressure Data, Analysis, and Model

[Fig. 10](#) provides an example of a water hammer signature that was induced at the end of a treatment stage when the injection rate was shut down rapidly. The x-axis is the time in seconds since the rate shutdown. The red series is the treating pressure; the green series is the rate.

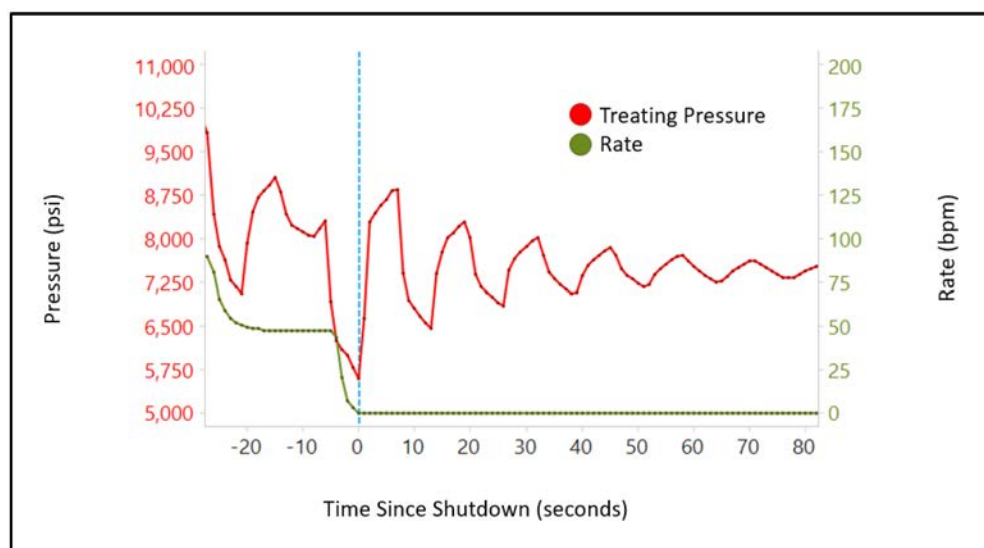


Figure 10—Water hammer example

In the company's completions database, there are over 1,200 wells with over 40,000 stages of one-second treatment data. The capability to iteratively develop and improve the analysis method/modeling and to

efficiently use the treatment data from the completions database facilitated our learning in respect to the shut-in process and the subsequent water hammer signature.

This section reviews the data considerations/requirements, the data analysis methods, and modeling.

Data Collection Frequency

Treatment pressure data is typically recorded at a frequency of 1 Hz (1 data point per second). A high frequency pressure gauge was used to determine if 1 Hz was an acceptable sampling frequency to adequately capture the characteristics of the water hammer that is induced by sharply reducing or terminating the treatment injection rate.

Data Collection Frequency Study using High Frequency Pressure Transducer Data. Pressure data was recorded at a sampling frequency of 50 Hz, and the resulting data was edited to lower sampling frequencies to compare the resulting quality of the water hammer signature. A key assumption for this exercise is that the specifications (e.g., accuracy, resolution, frequency response) for the high frequency pressure transducer would be similar to the pressure transducer being provided by the service company for the standard one-second frequency treatment data.

The water hammer pressure data shown in Fig. 11 is from a treatment with an average perforation depth of 17,370 ft MD with the original sampling frequency of 50 Hz and edited sampling frequencies of 2 Hz, 1 Hz, and 0.5 Hz.

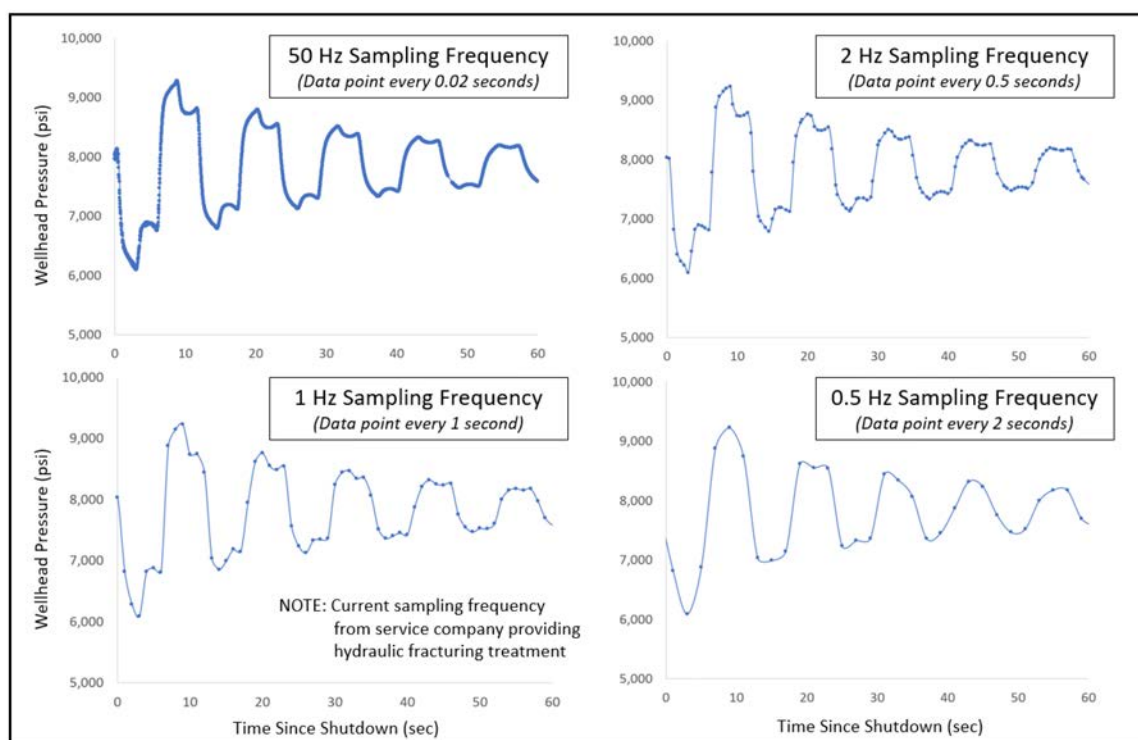


Figure 11—Water hammer data at various sampling frequencies (50, 2, 1, 0.5 Hz).

Sampling frequency observations on 50 Hz data and its edited low frequency data sets:

1. While the data recorded at 50 Hz shows more detail, the sampling frequency of 1 Hz captures the overall characteristics of the water hammer signature.
2. For this data set, 2 Hz was the lowest sampling frequency that appeared to show the full shape of the water hammer signature.
3. At 1 Hz and 0.5 Hz, the water hammer signature becomes much more smoothed with less character.

4. A sampling frequency of 1 Hz is adequate to characterize the water hammer period and decay rate. Higher sampling frequencies could be beneficial for performing more detailed analysis.

Comparison of Pressure Transducers (50 Hz versus 1 Hz Data from Service Company). For the same operation noted in the prior section, two transducers recorded wellhead pressure. One was the 50 Hz pressure transducer (non-standard for our normal hydraulic fracturing treatments); the other was the service company pressure transducer which provides one-second frac data (standard for our normal hydraulic fracturing treatments). The two data sets are compared in Fig. 12.

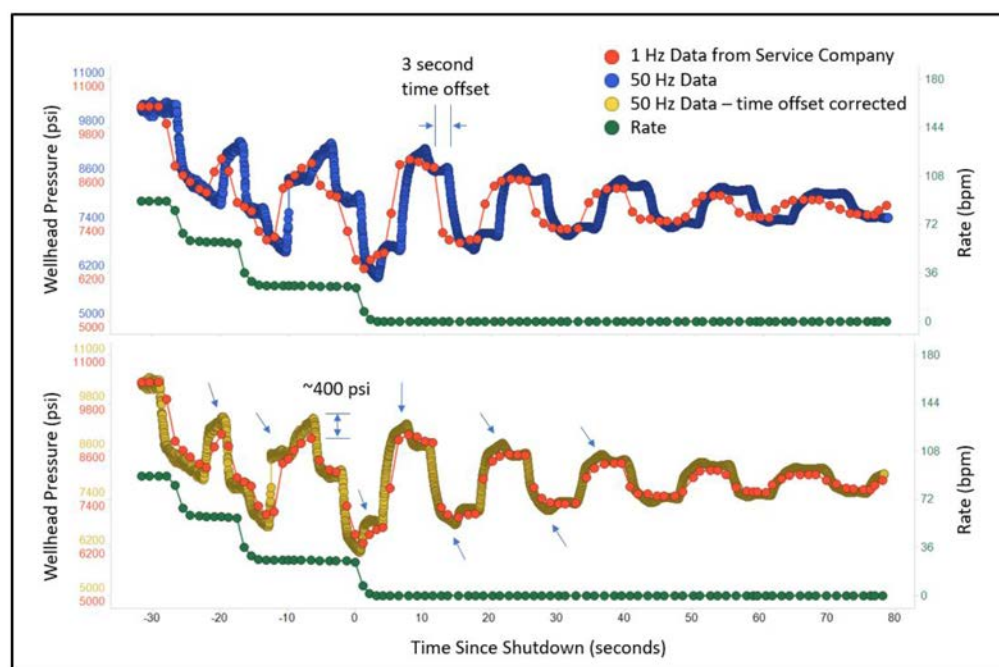


Figure 12—Comparison of high frequency versus one-hertz service company data.

The following are observations on the comparison of the 50 Hz pressure transducer versus the 1 Hz service company transducer:

1. The top chart in Fig. 12 compares the 50 Hz pressure data set (blue series) against the 1 Hz service company pressure data (red series). A 3 second offset between the two data sets was identified. On the bottom chart in Fig. 12, the 50 Hz pressure data was corrected with a time offset (yellow series) to line up with the 1 Hz service company pressure data. One data consideration/requirement is time synchronization of sensors during operations to minimize time offset corrections for analysis.
2. Overall, the water hammer signature corresponds between the two transducers. Both data sets have the same water hammer period and general shape.
3. The 1 Hz service company pressure data seems to be more smoothed (captures less of the water hammer character) and has lower peaks/higher troughs compared to the 50 Hz transducer. This is due to differences in pressure transducer specifications. The 50 Hz transducer has a faster frequency response to pressure changes compared to the 1 Hz service company transducer. For water hammer modeling and pressure matching, pressure transducer specifications should be considered.
4. The 1 Hz service company transducer measurement is adequate to characterize the water hammer period and decay rate. Higher sampling frequencies and improved pressure transducer specifications could be beneficial for performing more detailed analysis and water hammer modeling.

Data Collection Frequency for Shorter Wells. The water hammer period is a function of the speed of sound in fluid and the measured depth of the stage. On very shallow stages, the water hammer peaks will return to surface much faster and a sampling frequency of 1 Hz may not be adequate to fully capture the shape of the water hammer. The expected water hammer period can be calculated by using a rough estimate of 1 second per every 1,200 ft MD (4,000 m MD) of stage depth. It is recommended to use a sampling frequency that will collect at least 8 data points per water hammer period to ensure that the water hammer signature is adequately sampled.

The input assumptions for Table 1 is that the fluid speed of sound is ~5,000 ft/s (~1,500 m/s), and the boundary condition for the well is a closed inlet and a constant pressure outlet. For the boundary condition of closed inlet and closed outlet, the water hammer period is half of the values listed below.

Table 1—Perforation depth and water hammer period for closed inlet and constant pressure outlet.

Perforation depth (ft)	Perforation depth (m)	Water hammer period time (s)
5,000	1,524	4
10,000	3,049	8
15,000	4,573	12
20,000	6,098	16

Data Issues and Requirements

Over the course of evaluating shut-in pressure data, various data issues have been encountered that result in analysis issues. From over 15,000 stages analyzed from hydraulic fracturing treatments in Texas, South America, and Canada, approximately 20% of stages had confirmed data quality issues. The following are the data quality issues encountered:

1. Data stops before water hammer ends
2. Incorrect representation of wellhead pressure
3. False injection rates
4. Smoothed data
5. Data accuracy

There are operational considerations and data requirements that can be implemented to reduce these data quality issues. Data quality requirements can further be referenced in the Data Quality Assurance Contract Addendum posted on the Operators Group for Data Quality website (<http://www.OGDQ.org>).

Currently, the predominant method of acquiring treatment data from service companies is through CSV (comma-separated values) files. After the end of a treatment stage, an engineer from the treatment service company provides a post job report and a CSV file containing 1 Hz data for the hydraulic fracturing treatment. The data includes time, pressure, and rate. For the treatment stage, the engineer manually selects the start and end time of the data to be exported into the CSV file.

As third-party aggregation services further develop and improve in the completions space, these same data quality issues will need to be reviewed and addressed.

Issue #1 – Data Stops before Water Hammer Ends. Fig. 13 shows an example of a treatment stage where only 10 seconds of data was provided for the shut-in period. This is insufficient time to evaluate the water hammer signature.

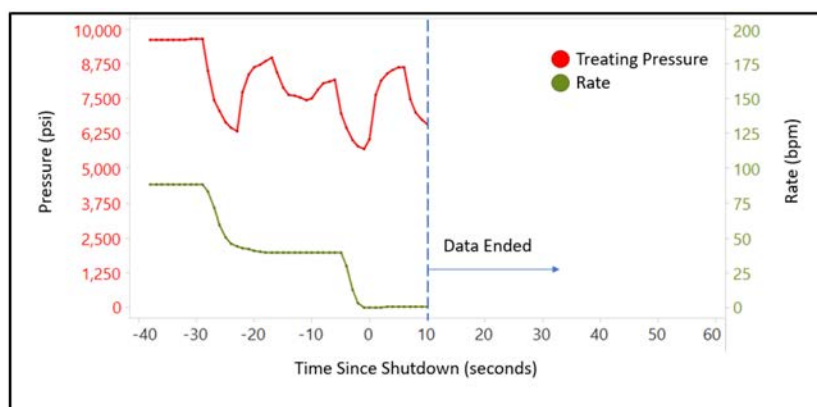


Figure 13—Provided data stops before water hammer ends.

Issue #2 – Incorrect Representation of Wellhead Pressure. Below are examples where the Treating Pressure transducer does not represent the wellhead pressure.

Scenario #1. Fig. 14 shows the equipment configuration where a valve is shut isolating the treating pressure transducer from the wellhead. When the valve is closed, Pressure Sensor #1 measures the pressure in the surface lines upstream of the valve (blue colored line) and not in the surface lines downstream of the valve (black colored line) which would be the wellhead pressure.

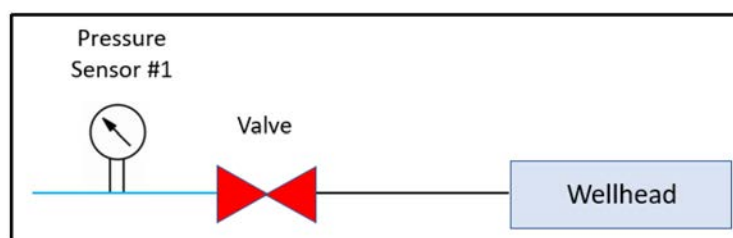


Figure 14—Configuration of Pressure Transducer, Valve and Wellhead

Three examples of Scenario 1 are provided in Fig. 15. Pressure Sensor #1 (in Fig. 14) is providing the Treating Pressure noted in Fig. 15. The dashed blue vertical line represents the time at which the valve was closed. Once the valve is closed, the Treating Pressure no longer represents the wellhead pressure.

1. For first and second example, after the valve was closed, the pressure was not bled off immediately. The pressure trend between the valve closure and the pressure bleed off represents the pressure in the surface line upstream of the closed valve. This pressure trend does not represent the wellhead pressure. A flat pressure trend means the pressure is holding, a declining pressure trend means there is a loss of pressure (like a leak), and an inclining pressure trend means there is an increase in pressure (due to pumping or temperature fluid expansion).
2. For the third example, after the valve was closed, the pressure in the upstream surface lines was bled off immediately.

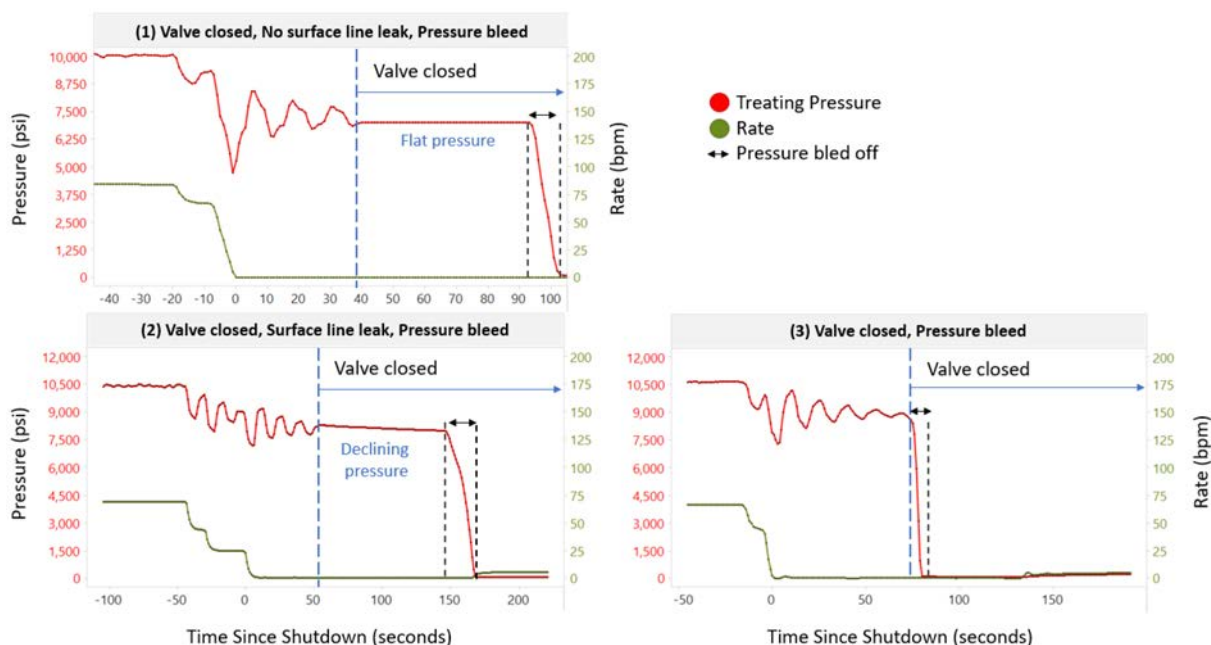


Figure 15—Incorrect representation of wellhead pressure - Valve closed

Scenario #2. Multiple pressure transducers may be installed in the surface treating lines. The service company engineer selects which are to be viewed and recorded in the Treating Pressure channel during the treatment operation. If the pressure sensor selected as the Treating Pressure channel is located on the injection pump side and upstream of check valves, and the injection rate is terminated, the pressure transducer could become isolated from the wellhead pressure.

The basic configuration is shown in Fig. 16. Pressure Sensor #1 and Pressure Sensor #2 are two transducers on the surface line from which the service company engineer can select to represent the Treating Pressure. Pressure Sensor #1 is upstream of the check valve; Pressure Sensor #2 is downstream of the check valve. Check valves allow flow in one direction, from left to right as indicated by the arrow on the check valve symbol. If the pressure is greater downstream of the check valve than upstream, the check valve will prevent flow going back upstream thereby isolating Pressure Sensor #1 from Pressure Sensor #2. Afterward, the two pressure sensors will have different readings.

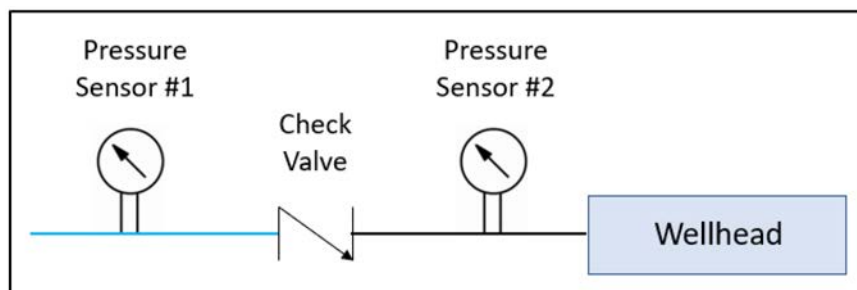


Figure 16—Configuration of Pressure Transducer, Check Valve and Wellhead

The example shown in Fig. 17 indicates that initially Pressure Sensor #1 was selected as the Treating Pressure channel. The Treating Pressure channel properly represented the wellhead pressure until the rate dropped to zero. At this time, wellhead pressure dropped due to the Joukowski effect. Eventually the wellhead pressure increased due to rebound of the water hammer pulse. The associated reverse flow up the wellbore caused the check valve to close, resulting in the pressure upstream of the check valve being lower than the pressure downstream of the check valve. At this point, Pressure Sensor #1 was isolated from

Pressure Sensor #2 by the check valve. Around 10 seconds into the shutdown, the service company engineer recognized the wellhead pressure was not reading correctly, then switched to Pressure Sensor #2 for the Treating Pressure channel. An estimate of the missing water hammer pressure is drawn in blue.

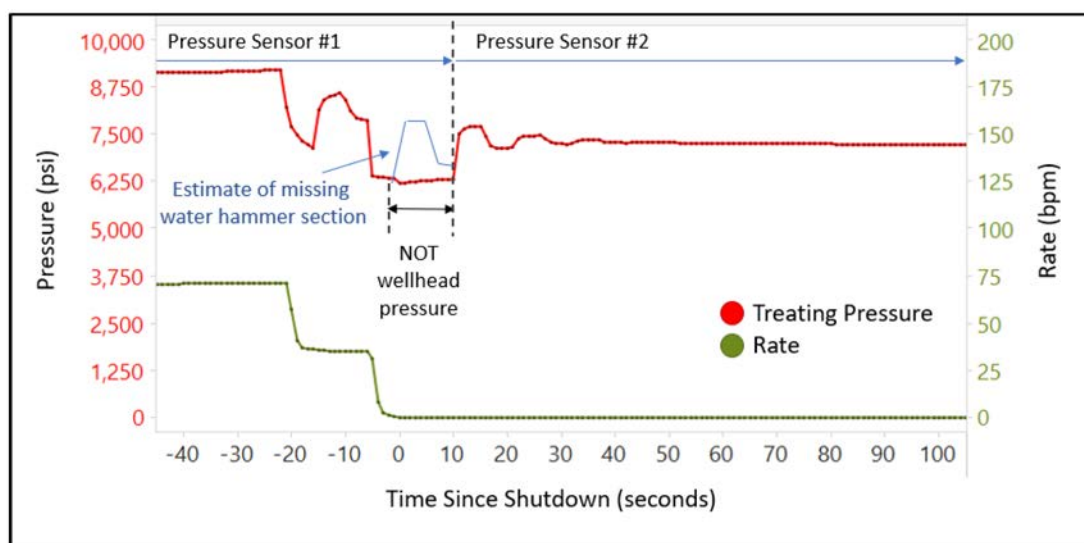


Figure 17—Incorrect transducer selected

Issue #3 – False injection rates. Fig. 18 shows an example where there is an indication of rate during the shut-in period. As there is no associated pressure increase related to the rate, this is considered a "false injection rate" as this rate is not representative of rate being injected down the well but indicates pumping for a surface only operation. The issue with the false injection rate is that rate is used to identify the shut-in period. False injection rates may result in the incorrect identification of the shut-in period.

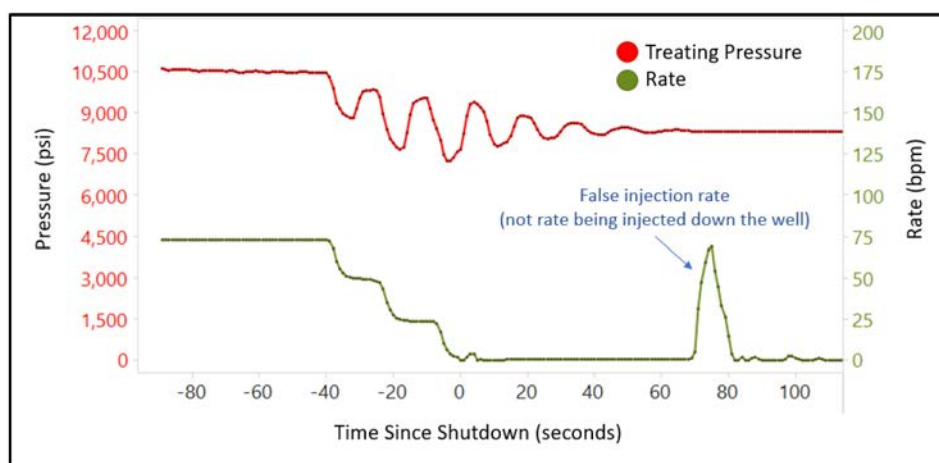


Figure 18—False injection rate

Issue #4 – Smoothed data. Treatment data may not be instantaneous values but be smoothed by averaging over a set amount of time (e.g., over 10 seconds). This results in difficulty in connecting pressure with rate changes and to identify events such as the start of the shut-in period. An example of the injection rate being smoothed by averaging it over a 40-50 second period is shown in Fig. 19. Issues related to smoothing of rate are:

- If shutdown is identified by using a rate threshold (like 0.1 barrels per minute), the start of shutdown may be delayed by 40 seconds. Multiple periods of the water hammer may not be identified correctly.
- Also due to the smoothing, it is difficult to identify the distinct step-down rates.

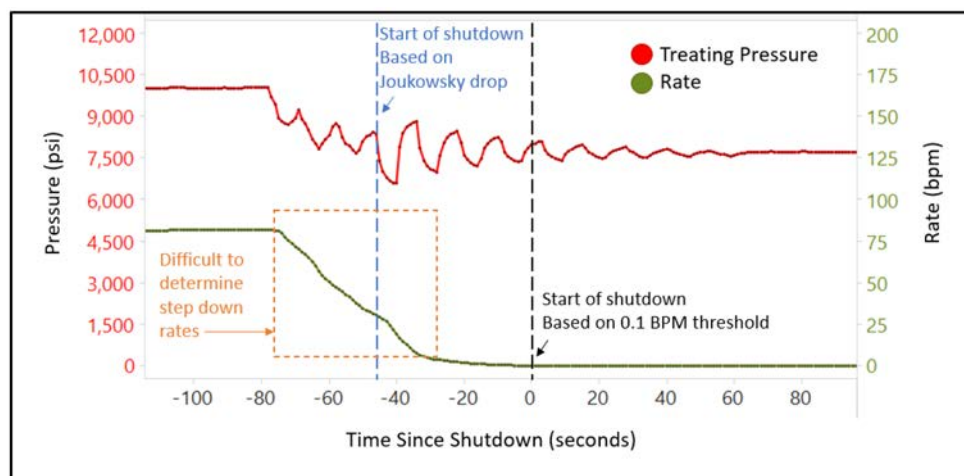


Figure 19—Smoothed injection rate

With smoothing of pressure data, the water hammer signature will be delayed in time and will lose its character.

Issue #5 – Data accuracy. In the example presented in Fig. 12, two pressure gauges (50 Hz and 1 Hz Service Company) matched overall in respect to the water hammer signature (same water hammer period and general shape) and the average pressure (minimal offset). This was a positive observation for these two pressure gauges. A comparison of two gauges on a different treatment is shown in Fig. 20. One was a recently calibrated memory gauge (blue series); the other was the service company gauge (red series). Although the pressure of the two gauges have similar trends, there is a pressure offset between the gauges of about 120 psi as determined during the shut-in period.

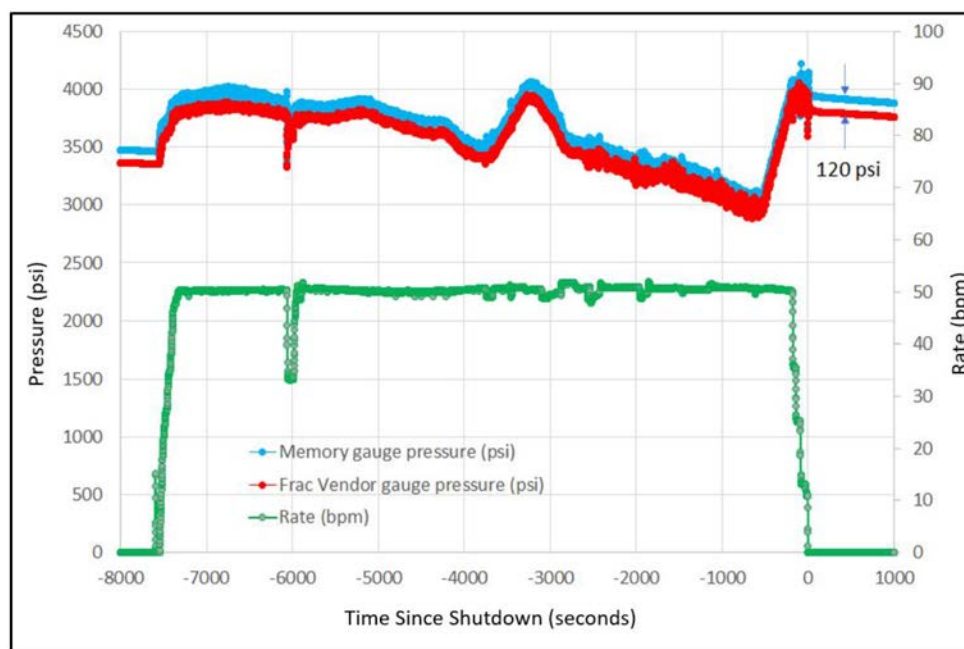


Figure 20—Comparison of memory gauge versus service company gauge

Data for the step-down and shut-in part of this treatment stage is expanded in Fig. 21. At the stepped down injection rate of ~12 bbl/min, the service company gauge exhibited an erratic pressure pattern rather than the expected decaying pattern for the induced water hammer. For the shut-in period, the two gauges had the general trend. The service company gauge did not capture accurately the detail of the water hammer signature compared to the memory gauge.

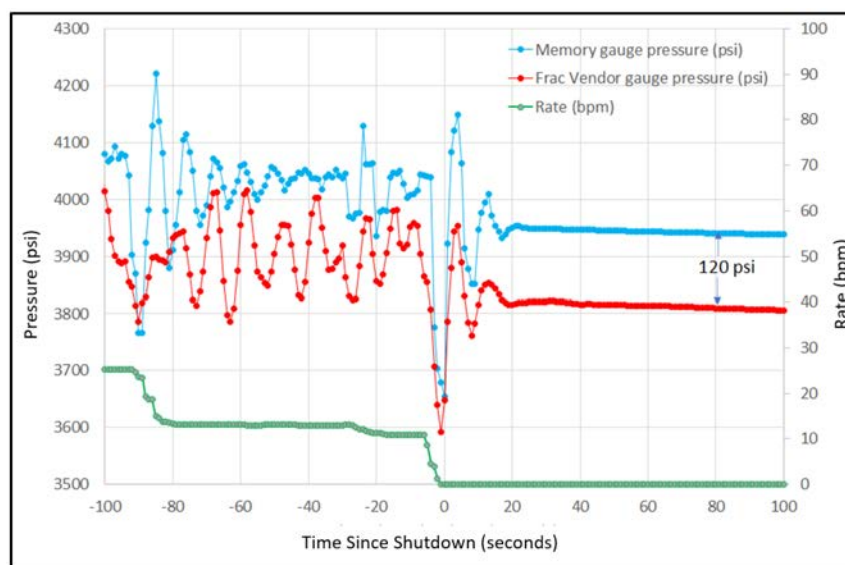


Figure 21—Comparison of memory gauge versus service company gauge

Fig. 22 shows data from the same service company gauge compared against two other pressure gauges (piezo resistive strain gauge, dual quartz gauge). The application was a diagnostic fracture injection test (DFIT) which requires high accuracy and resolution pressure data. The service company gauge registered false pressure drops and spikes.

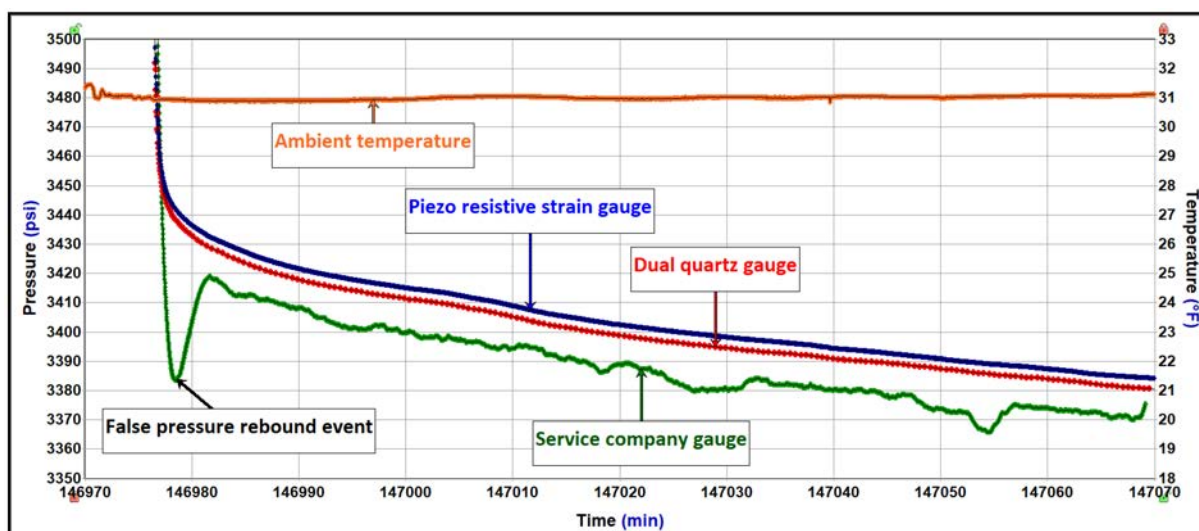


Figure 22—Comparison of memory gauge versus service company gauge, post injection shut-in period

In respect to accuracy, there may be:

1. a measurement offset (120 psi offset)
2. a device artifact/issue (false pressure drops and spikes)
3. a measurement responsiveness difference (difference in capturing slight characteristics of the water hammer)

Depending upon the application, instrumentation specifications should be considered. For the case of the service company gauge (Fig. 20–22), this gauge is adequate for overall pressure trends but is not suited for water hammer analysis or more precise pressure analysis (ex: DFIT).

Corrections for Data Quality Issues

There are three levels of actions to address the noted data quality issues.

1. File corrections:
 - a. Request that the service company provide a corrected CSV file by re-exporting the treatment stage data and include more shut-in data. This will correct situations where more shut-in data was recorded, but the frac engineer did not select sufficient shut-in data for the CSV file.
 - b. Request that the service company provide a corrected CSV file by re-exporting the treatment stage data and correcting the Treating Pressure channel to the appropriate sensor.
 - c. Manually correct the received CSV file to remove the false injection rates.
2. Algorithm corrections/improvements:
 - a. Develop algorithms to address smoothed injection rates for the identification of the shut-in period.
 - b. Develop algorithms to address false injection rates for the identification of the shut-in period.
3. Frac data requirements:
 - a. A minimum of 3 minutes is required for the shut-in period. This data requirement may conflict with goals for reducing time between operations. The operator will need to determine the priority of the requirements.
 - b. Installation of a pressure gauge to record wellhead pressure downstream of valves used to isolate multiple wells being treated sequentially. This will allow sufficient data to be acquired

without delaying sequencing operations. The continuous recording of wellhead pressures can also facilitate data acquisition.

- c. Request service company to ensure that all injection rate channels accurately reflect what is being injected into the well. This may require a process to zero-out the injection rate during the shut-in period.
- d. Set instrumentation and data collection system requirements on time synchronization, reading accuracy, reading resolution, data collection frequency, and data transformations (instantaneous versus smoothed readings).

Water Hammer Analysis

The water hammer analysis consists of 4 parts:

1. Identification of the shut-in period.
2. Identification of water hammer peaks and troughs.
3. Calculation of water hammer period and the number of periods.
4. Calculation of water hammer decay rate (based on peak and trough pressure differences).

See Fig. 23 for the parts of the water hammer nomenclature.

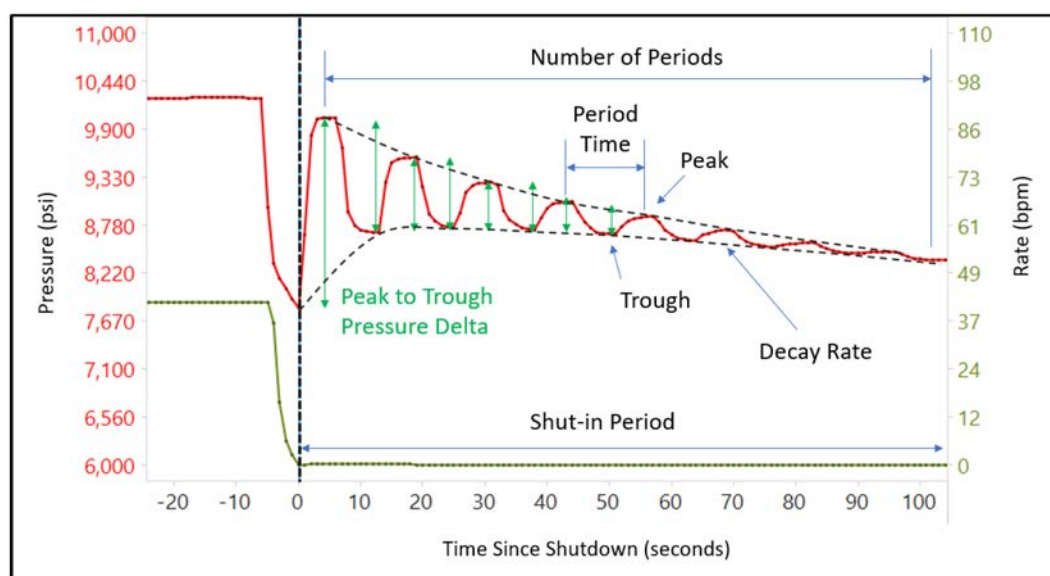


Figure 23—Water hammer nomenclature

The shut-in period is identified using the total injection rate. At the end of the treatment stage, the start of the shut-in period is based on a rate threshold considered to be zero rate. Due to potential noise in the rate sensor, values of zero may not be recorded so data is reviewed for an appropriate rate threshold (e.g., reading less than 0.1 barrels per minute is considered zero). As noted in the Data Issues and Requirements section, additional conditions/adjustments are required to handle false injection rates or smoothed rate data.

The next step is to identify the peaks and troughs of the water hammer signature as shown in Fig. 24. Peaks and troughs are identified with yellow vertical lines. A simple algorithm to select peaks and troughs is the following.

- A point is a peak if the adjacent points on either side of it have values lower than it.
- A point is a trough if the adjacent points on either side of it have values higher than it.

For water hammers of this shape, this simple algorithm can be used to identify the peaks and troughs, and their values and respective times.

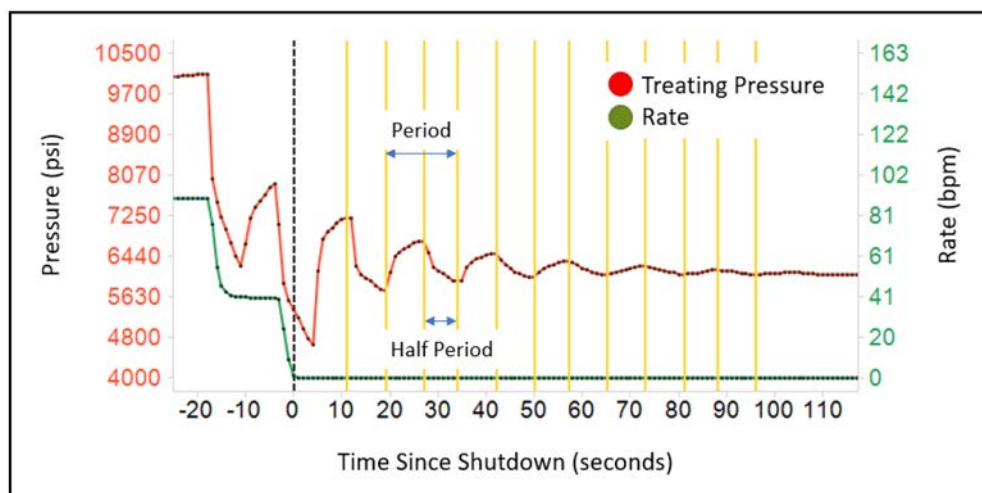


Figure 24—Picking peaks and troughs

Due to varying water hammer shapes or potentially noisy pressure data, the simple peak/trough algorithm is not sufficient in all cases. This is exemplified in Fig. 25. Additional conditions and/or signal processing is required to handle more water hammer cases automatically.

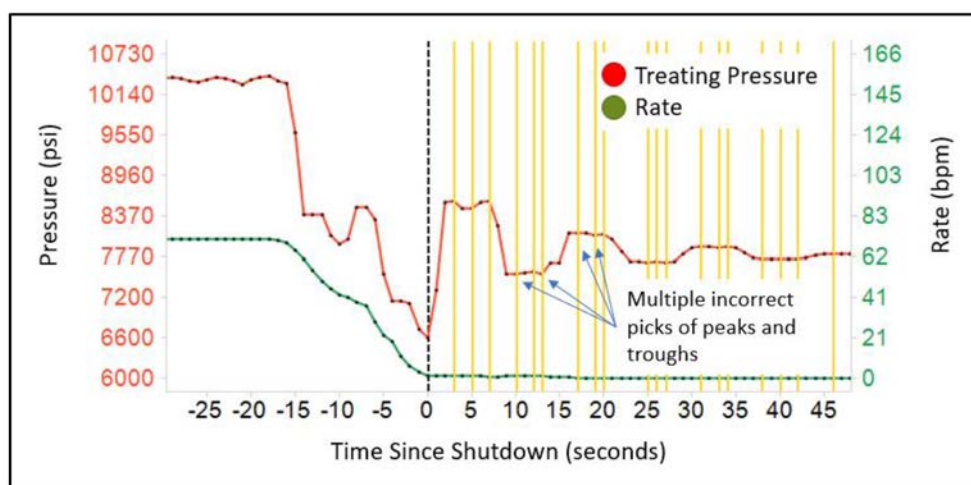


Figure 25—Picking incorrect peaks and troughs

The next step is to calculate period and the number of periods. A period is from peak to peak or trough to trough. A half period is from peak to trough or trough to peak. As shown in Fig. 24, once the water hammer signature decays to a point where the difference between peak and trough pressures are below a specified differential pressure threshold, half period are no longer identifiable. The values tabulated for the case depicted in Fig. 24 are shown in Table 2.

Table 2—Water hammer calculations for the case depicted in Fig. 24.

# Half Periods	# Periods	Start Time (sec)	Half Period Δ Time (sec)	Peak/Trough Value (psi)	Peak and Trough Δ Pressure (psi)
0	0	4		4,671	
1	0.5	11	7	7,214	2,543
2	1	19	8	5,751	1,463
3	1.5	27	8	6,734	983
4	2	34	7	5,945	789
5	2.5	42	8	6,494	549
6	3	50	8	6,022	472
7	3.5	57	7	6,345	323
8	4	65	8	6,059	287
9	4.5	73	8	6,239	180
10	5	81	8	6,076	163
11	5.5	88	7	6,161	85
12	6	96	8	6,079	82
Average =			7.7		

The total number of periods is the count of half periods divided by two. For this case, there are 12 half periods, so the number of periods is 6. To calculate the period, the differential time between the half periods are calculated. The average of the half period differential time is calculated. For this case, the average is 7.7 seconds for the half periods. The period is twice the half period average which is 15.4 seconds. With more half periods, the average period becomes more accurate as issues with properly picking the start and end times of the half periods get averaged out.

The last step is to calculate the water hammer decay rate, as shown in Fig. 26.

- The log of peak and trough differential pressure plotted versus shut-in time (seconds) is linear. The decay rate is represented by an exponential decay.
- The decay rate for this case is -0.039 .
- The R^2 value of 0.986 indicates a good correlation.

In the development of the decay rate function, the exponential relationship was found to provide the best correlation.

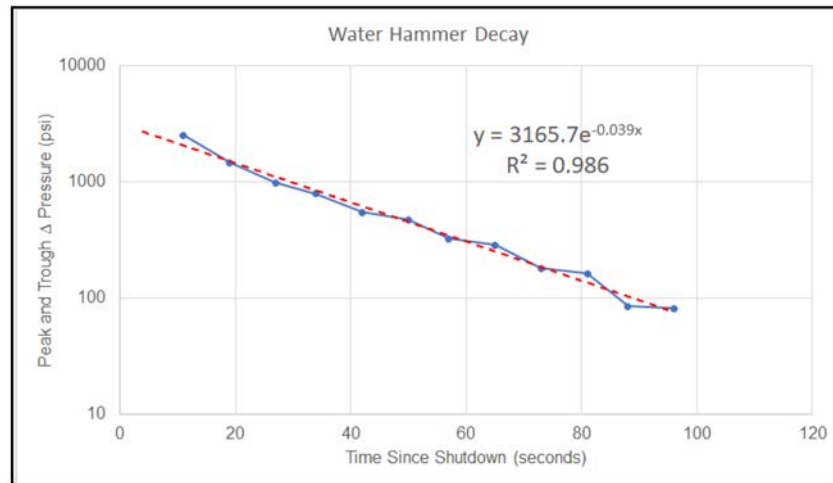


Figure 26—Water hammer decay for the case depicted in Fig. 24.

This measurement provides an indication of the friction of the system that may facilitate at least a qualitative understanding of the hydraulic fracture network and its connection with the wellbore.

Water Hammer Model

The water hammer numerical method used in this study is described as follows. A staggered-grid method is used, with the one-dimensional momentum equation solved on the primary grid in Fig. 27 and the mass conservation equation solved on the staggered grid in Fig. 28. The nomenclature for the below figures is:

A_k : cross-sectional area in momentum grid at position k

y_k : elevation in momentum grid at position k

z_k : distance in momentum grid at position k

$\bar{Z}_k$: distance in mass grid at position k

ρ_k : density in momentum grid at position k

$\bar{\rho}_k$: density in mass grid at position k

P_k : pressure at position k

Δz_k : length of momentum grid at position k

$\Delta \bar{z}_k$: length of in mass grid at position k

$a\Delta z_k$: volume of in mass grid at position k

$\dot{m}_k$: mass rate at position k

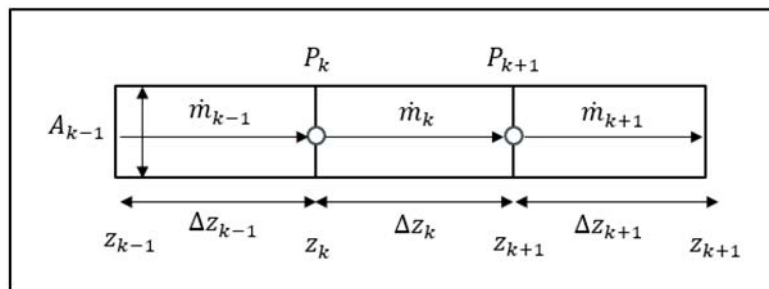


Figure 27—Grid for one dimensional momentum equation

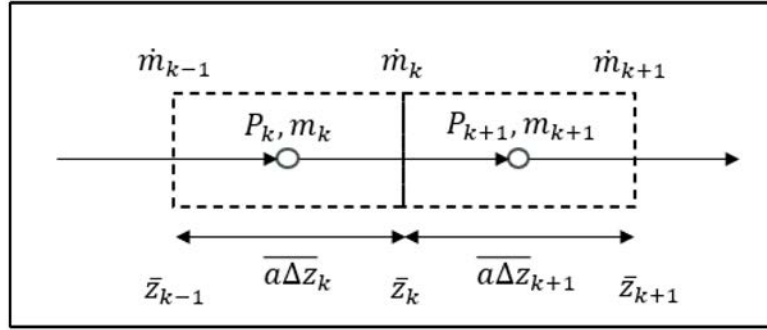


Figure 28—Grid for mass conservation equation

The velocities and mass rates are stored at the cell centers of the primary grid z_k , and the pressures, temperatures, fluid properties, and masses are stored on the staggered grid $\bar{z}_k$, i.e., at the boundaries of the primary grid.

The volume of the staggered cell at position k , $a\bar{\Delta}z_k$, is given by:

$$a\bar{\Delta}z_k = \frac{A_k \Delta z_k + A_{k+1} \Delta z_{k+1}}{2} \quad (4)$$

where A_k is the cross-sectional area of cell k , Δz_k is the length of cell k .

The momentum conservation equation is written as follows:

$$\Delta z_k \frac{dm_k}{dt} = (um)_{in} - (um)_{out} + \sum F_k \quad (5)$$

where the spatial momentum terms are given by a first-order upwind scheme

$$(um)_{in} = \max(u_{k-1}, 0)m_{k-1} - \min(u_{k+1}, 0)m_{k+1}; (um)_{out} = |u_k|m_k \quad (6)$$

and the forces acting on the fluid are given by:

$$\sum F_k = (P_k - P_{k-1})A_k - \frac{f_k |u_k| \Delta z_k}{2D_k} m_k + (y_k - y_{k-1}) \bar{\rho}_k g A_k \quad (7)$$

where the first term is the pressure force acting on cell k , the second term is the frictional force acting on cell k , and the third term is the gravitational force acting on cell k . The density in this cell is given by:

$$\bar{\rho}_k = \frac{\rho_k A_k \Delta z_k + \rho_{k+1} A_{k+1} \Delta z_{k+1}}{A_k \Delta z_k + A_{k+1} \Delta z_{k+1}} \quad (8)$$

The mass conservation equation is written:

$$\frac{dm_k}{dt} = \dot{m}_{k-1} - \dot{m}_k \quad (9)$$

This equation can be re-written:

$$\frac{a\bar{\Delta}z_k}{c_k^2} \frac{dP_k}{dt} = \dot{m}_{k-1} - \dot{m}_k \quad (10)$$

where c_k is the speed of sound at the boundary of cell k . The mass equation is then solved as follows:

$$P_k = P_k^o + \frac{c_k^2 \Delta t}{a\bar{\Delta}z_k} (\dot{m}_{k-1} - \dot{m}_k) \quad (11)$$

P_k^o is the pressure at position k at the beginning of the time step and P_k is the pressure at the end of the time step. Using this relationship, derived from the mass conservation equation, the pressure term in the momentum equation is then replaced as follows:

$$(P_k - P_{k+1})A_k = (P_k^o - P_{k+1}^o)A_k + \frac{c_k^2 \Delta t}{a \Delta z_k} (\dot{m}_{k-1} - \dot{m}_k) - \frac{c_{k+1}^2 \Delta t}{a \Delta z_{k+1}} (\dot{m}_k - \dot{m}_{k+1}) \quad (12)$$

This gives the momentum equation of the form:

$$\Delta z_k \frac{d\dot{m}_k}{dt} = a_k \dot{m}_{k-1} + b_k \dot{m}_k + c_k \dot{m}_{k+1} + s_k \quad (13)$$

These equations produce a tri-diagonal matrix which can be inverted directly, without iteration. This matrix allows for an implicit solution of the mass rate vector, $\dot{m}_k$. Once the mass rates at the new time step are determined, they are used to update the pressure vector, P_k .

The native speed of sound in a material is related to its density and bulk modulus according to the equation

$$K = \rho C_o^2 \quad (14)$$

where

- ρ is the fluid density
- C_o is the native speed of sound of the liquid
- K is the bulk modulus of the liquid

In addition, the speed C of a pressure impulse in a pipe must be modified to accommodate

- Pipe geometry
 - Inner diameter, D
 - Wall thickness, T
- Pipe material
 - Young's modulus E
 - Poisson's ratio, ν
 - Nature of anchoring, ψ

The modified speed of sound C in a pipe is given by:

$$\left(\frac{C_o}{C}\right)^2 = 1 + \frac{\psi DK}{TE} \quad (15)$$

where, for a line anchored throughout (casing cemented in):

$$\psi = \frac{2T}{D}(1+\nu) + \frac{D}{D+T}(1-\nu^2) \quad (16)$$

The water hammer model is completely general, and can accommodate:

- Complex well geometries, including changing diameter
- Changing properties through the well, including density, speed of sound, and viscosity
- Influence of drag reduction chemical on friction factor
- Pressure drop across the perforations (using a simplified choke model)
- Bulk modulus of the well casing (including the effects of the steel and cement)

In the event that there is some gas entrained in the liquid, the native speed of sound C_0 must be modified still further, as even a small amount of gas will have a very large impact on the speed of sound in the fluid. For a gas-liquid flow, the bulk modulus of the fluid is given by:

$$\frac{1}{K} = \frac{H_L}{\rho_L C_L^2} + \frac{H_G}{\rho_G C_G^2} \quad (17)$$

where H_L and H_G are the liquid and gas volume fractions, and C_L and C_G are the speed of sound in liquid and gas, respectively.

The model has been tested against dozens of wells and hundreds of stages, with good fit to data, sometimes including small details in the pressure signature. Fig. 29 shows a comparison of the wellhead pressure during ramp-down of the slurry injection rate predicted by the water hammer model and measured in the field.

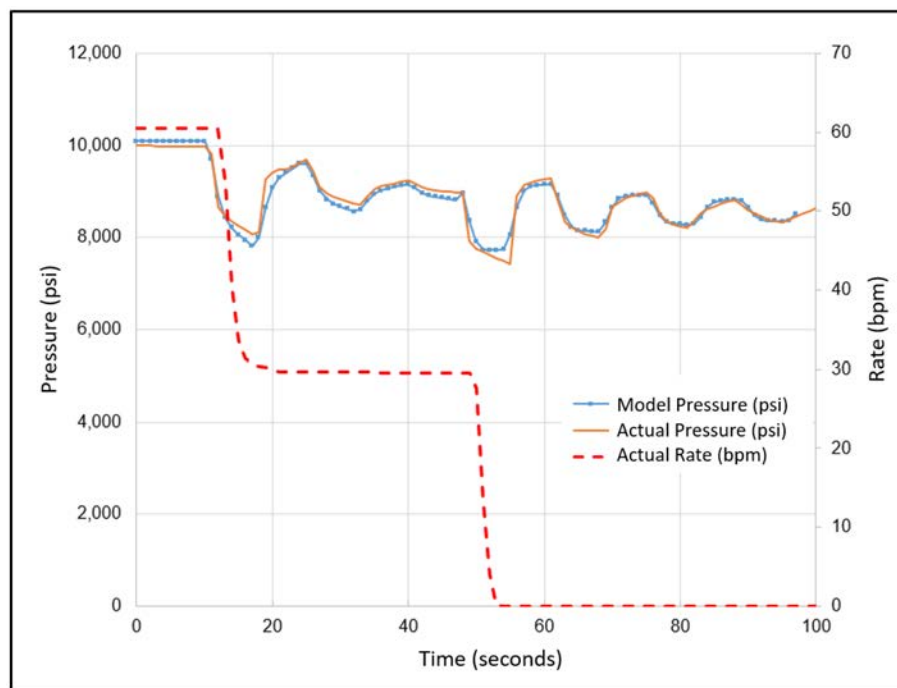


Figure 29—Fit of water hammer model to field data

This current developed model incorporates wellbore properties including perforations but does not incorporate the fracture network. The model provides the influence of the wellbore to the water hammer signature. Differences between the model and the actual field data can provide insight into the influence of the fracture network on the water hammer signature.

Model Comparison to Actual Data. The following are the main levers for history matching the water hammer signature with the model as demonstrated in Fig. 30.

1. Outlet constant pressure condition is set to the ISIP and used to match wellhead pressure.
2. Friction is adjusted to affect the decay rate
 - a. Drag reduction factor, which affects the pipe friction
 - b. Perforation friction
3. Well length plus excess length. The excess length is added to increase the period. This additional length may be an indicator of the extent of the fracture network or fluid/casing property anomalies that reduce the fluid speed of sound.

The parameters are tuned for a stage and applied to following stages.

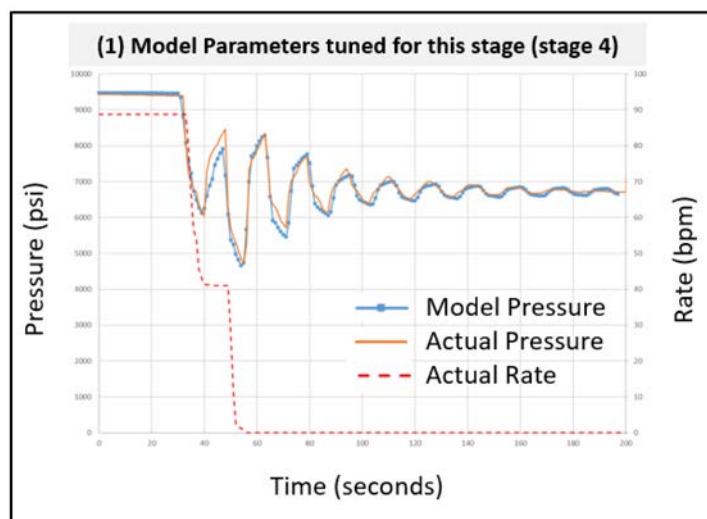


Figure 30—Model tuning

Observations on applied tuned model parameters to other stages, as exemplified in Fig. 31.

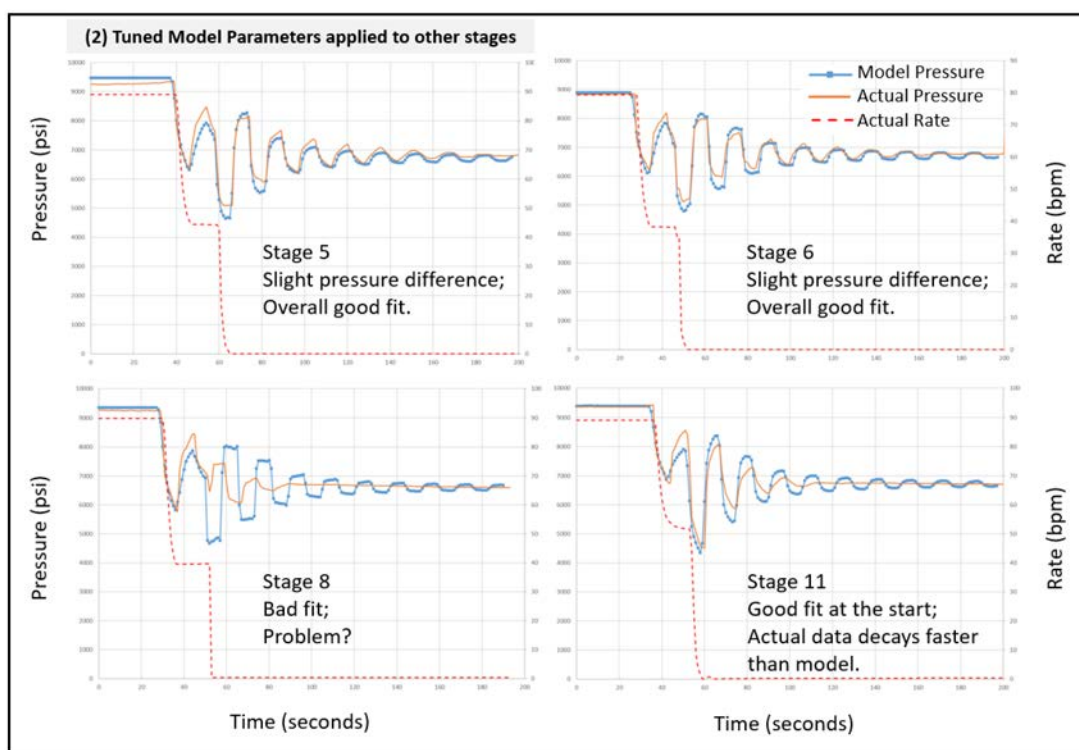


Figure 31—Tuned model applied to other stages

- The tuned model parameters provide a good match between the model and actual field pressure for stage 5 and 6. This is an indication that stages 4-6 are similar in respect to the wellbore, fluids, and fracture network created.
- For Stage 8, the actual water hammer signature does not match the model at the start of the shut-in. The difference may be due to data collection issues and/or friction changes which are causing the dampening.

- For Stage 11, the fit is good at the start; however, for this stage, the actual water hammer signature is dampening quicker than the model. It is uncertain whether the additional friction is due to fluid changes and/or the creation of a larger fracture network.

Effects on Water Hammer Signature

As the injection pumps reduce or terminate rate near the end of the treatment, a water hammer pressure signature will be created at the pump discharge. The nature of this signature depends on:

- fluid speed of sound in casing
- friction in the wellbore/fracture system
- the boundary condition at the top and bottom of the well.
- the nature of the step-down (i.e., step-down rate change and duration)

The following water hammer model sensitivity studies were conducted to understand the effect of key parameters on the water hammer signature.

Fluid properties

In order from highest to lowest effect, the following fluid properties affect the water hammer signature.

- fluid speed of sound in casing (affects the period).
- turbulence suppression - friction reducers in the fluid affect the development of turbulent eddy currents which thereby reduce friction (affects the water hammer decay rate).
- shear behavior – can affect friction reducer performance and/or actual fluid in respect to its gelling tendency (affects the decay rate).
- viscosity – increase in viscosity increases friction (affects the decay rate).
- density - impacts the speed of sound (affects the period).

The fluid speed of sound in casing is affected by fluid properties (e.g., density, bulk modulus) and casing properties (Poisson's ratio, bulk modulus, internal diameter, wall thickness). The fluid speed of sound affects the period.

$$C = \frac{1}{\sqrt{\rho * \frac{1}{K} + \frac{d^3(1-\nu^2)}{E * t}}} \quad (18)$$

C = fluid speed of sound in casing, m/s

ρ = fluid density, kg/m³

K = fluid bulk modulus, Pa

ν = casing Poisson ratio

d = casing internal diameter, m

E = casing bulk modulus, Pa

t = casing wall thickness, m

An example of the injection rate being stepped down in multiple steps is shown in [Fig. 32](#). The frac fluid was guar-borate crosslinked gel, and the flush fluid in the wellbore was low viscosity slick water. For the rate step-down to 13 bbl/min, the water hammer signature had up to 6 periods while following the shutdown when injection was completely terminated, there were only 2 periods.

- At 13 bbl/min, the fluid was subject to shear forces thereby reducing its viscosity and tendency to form a rigid gel structure (characterized by a high yield point). Having greater fluidity resulted in less friction and less decay of the water hammer signature.
- At shutdown (0 bbl/min), there was no shear forces induced by pumping acting on the fluid. The fluid thickened which resulted in more friction and a quicker decay of the water hammer signature.

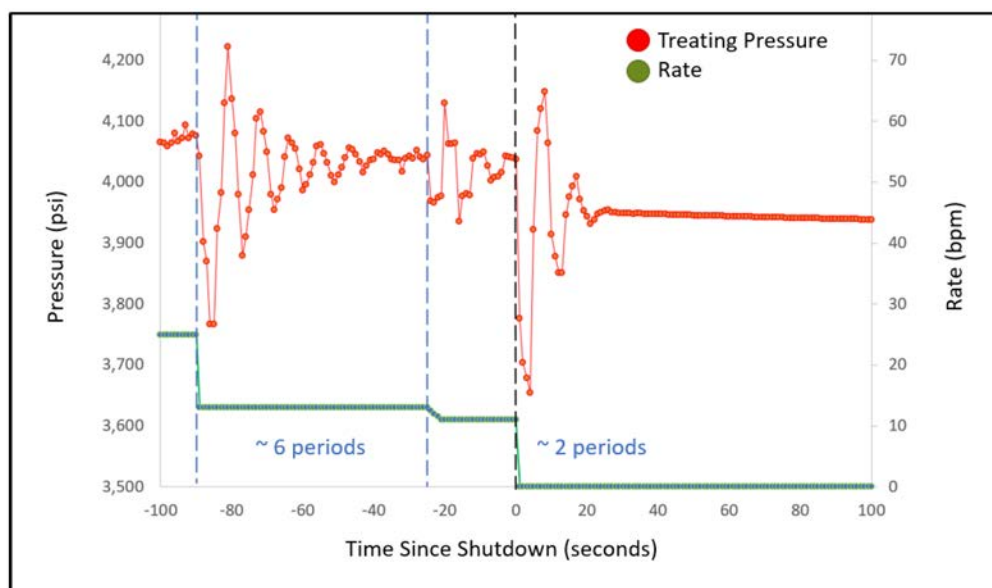


Figure 32—Effect of eliminating shear on fluid viscosity, yield point and water hammer signature

Step-down rate change and duration

The water hammer model described in the above section was used to perform a sensitivity analysis on the effect of step-down rate change and duration on the water hammer signature.

The concept of step-down rate change and duration is outlined in Fig. 33. The green series is the injection rate.

- Near the end of the hydraulic fracturing treatment, the injection rate is ~75 bbl/min.
- The injection rate is reduced by 35 bbl/min, from 75 bbl/min to 40 bbl/min.
- The injection rate is held at 40 bbl/min for a duration of about 30 seconds.
- The injection is completely terminated as rate is reduced from 40 to 0 bbl/min.

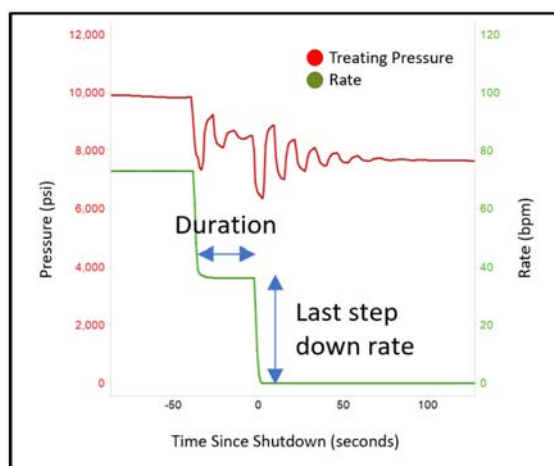


Figure 33—Illustration of step-down rate and duration

For the sensitivity cases below, the fixed model inputs are:

1. 20,000 ft from the wellhead to the perforations.
2. Fluid speed of sound through the wellbore is 5,000 ft/s.
3. Boundary conditions: closed inlet and constant pressure outlet, Boundary Condition Factor = 4
4. Injection rate prior to rate step-downs = 66 bbl/min.

With the above inputs, the calculated period is 16 seconds per Equation 3 ($4 \times 20,000 \text{ ft} / 5,000 \text{ ft/s} = 16 \text{ seconds}$).

Sensitivity Analysis #1: The results of a sensitivity analysis for three cases in which the initial rate is 66 bbl/min, the rate is reduced to 33 bbl/min with varied step-down duration time less than the period (15, 12, and 8 seconds), and then shut-in are shown in Fig. 34.

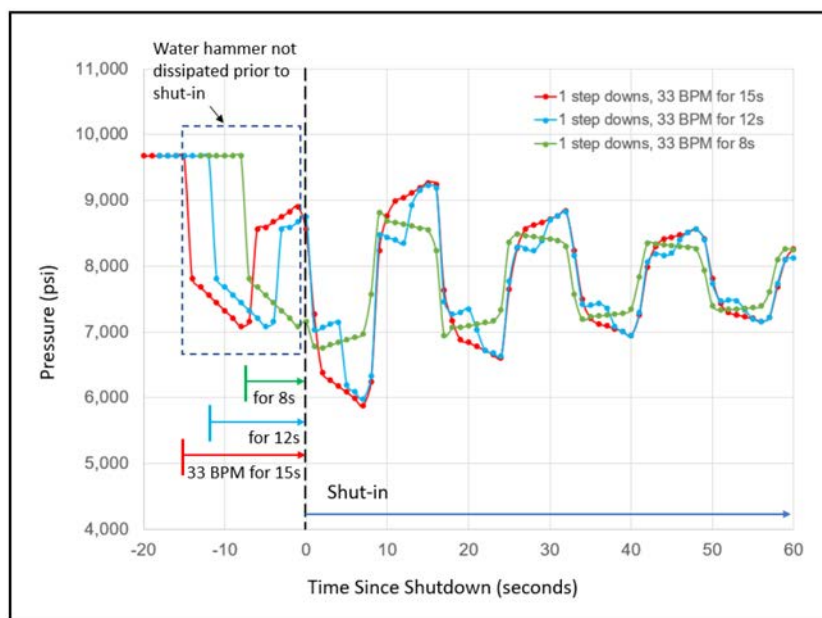


Figure 34—Model results, comparison of step-down durations when less than period.

Observations:

1. For step-down duration equal to 15 seconds, the peaks are showing an upward slope to the right.
2. For step-down duration equal to 12 seconds, the peaks are showing a half downward slope, then a half upward slope.
3. For step-down duration equal to 8 seconds, the peaks are showing a full downward slope.
4. With short step-down duration times, the water hammer signature induced by the first step-down does not have enough time to dissipate. The water hammer signature seen at shut-in is a combination of pressure wave remaining from the first step-down and the pressure wave created by the second step-down (shut-in).
5. A pressure superpositioning effect is seen with step-down durations less than the period resulting in the gradual change in slope from upward sloping to downward sloping. When the duration is half the period, the slope becomes completely downward sloping.
6. When the step-down duration equals half the period, this results in a 180° phase offset between the water hammer signature induced by the first and second step-downs. 180° phase offset means the peak of one pressure waveform coincides with the trough of the second pressure waveform.

The following are pressure-rate plots of actual treatments on the same well validating the modeling outcomes. For the treatment stage plotted in Fig. 35 (stage 23), the period was 13 seconds, calculated by measuring peak to peak. The last step-down rate was 30 bbl/min which was held for 12 seconds, close to the period. The peaks were upward sloping to the right.

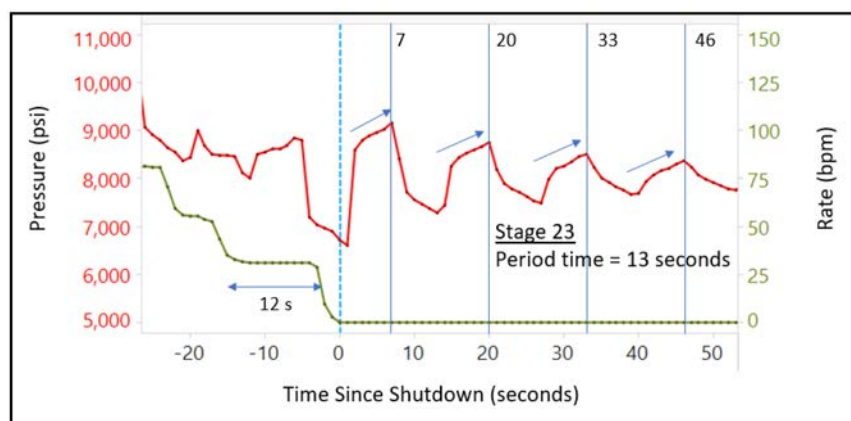


Figure 35—Actual data – upward slope related to step-down duration equal to period.

For the treatment stage plotted in Fig. 36 (stage 18), the period was 14 seconds. The last step-down rate was 28 bbl/min which was held for 4 seconds or less than half the period. The peaks are downward sloping to the right.

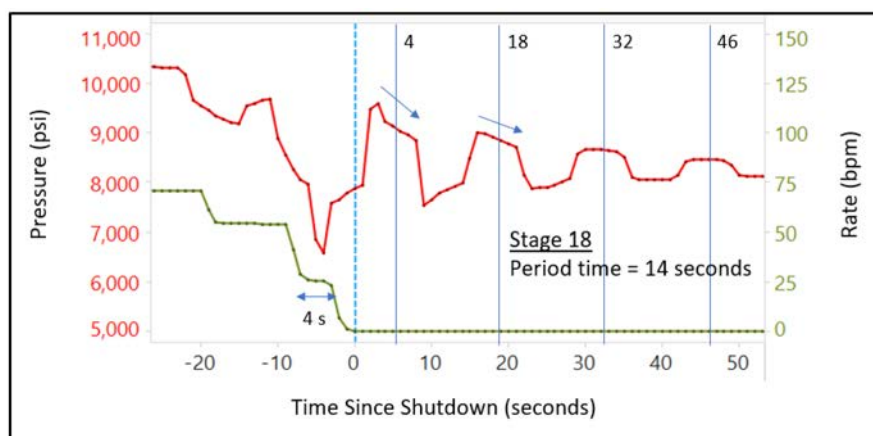


Figure 36—Actual data – downward slope related to step-down duration less than half the period.

This sensitivity analysis and actual treatment data observations indicate that the sloping nature of the water hammer signature is a function of the step-down duration time. It is recommended that step-down duration time is designed so that it is not less than the expected water hammer period.

Sensitivity Analysis #2: The results of a sensitivity analysis for two cases in which the initial rate is 66 bbl/min, the rate is reduced to 33 bbl/min with varied step-down duration times greater than the period (30 and 60 seconds), and then shut-in are shown in Fig. 37.

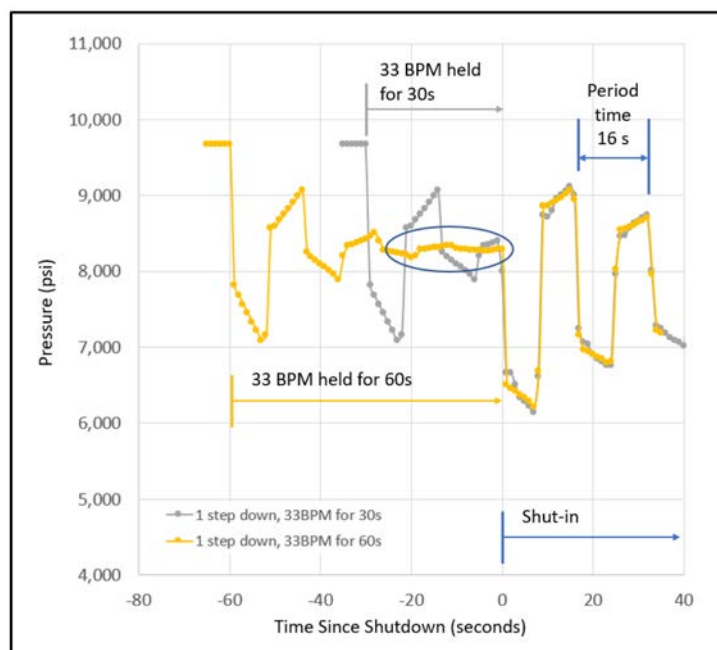


Figure 37—Model results, comparison of rate step-down duration when greater than period.

Observations:

1. For the simulation with a hold duration of 60 seconds, the water hammer signature is mostly dissipated around 30-40 seconds.
2. The final rate reduction (33 bbl/min to 0 bbl/min, shut-in) exhibited greater peak and trough pressure differentials than the first rate reduction (from 66 bbl/min to 33 bbl/min) even though both had the same 33 bbl/min rate reduction. The magnitude of the water hammer peaks and troughs are affected by continued fluid injection. For injection rate reductions of the same magnitude, zero rate during the

water hammer signature will have the greatest peaks and troughs while any rate greater than zero will reduce the water hammer signature. The higher the stabilized injection rate following the step-down, the greater the impact on water hammer signature reduction.

3. Reviewing the 30 second duration hold (case 1), there is a slight superpositioning effect seen during the zero rate section, but it is minimal when compared against the 60 second duration hold (case 2).

Based on the above results, it is recommended to hold constant the final rate step for at least 30 seconds to minimize water hammer superpositioning effects when operations require rate step-downs.

Sensitivity Analysis #3:. The following injection rate sensitivity was conducted to determine the effect of stepped down injection rate and the results are shown in Fig. 38. Maximum injection rate is established at 66 bbl/min. Rate is stepped down to various levels and held for 30 seconds. The rates modeled were 40, 35, 30, 25, 20 and 15 bbl/min. Injection rate is finally terminated, dropping to zero.

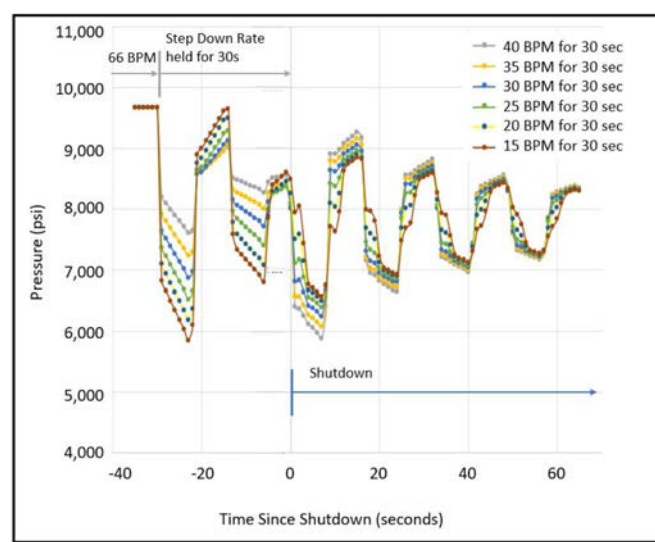


Figure 38—Model results - comparison of variable step-down rates, each held for 30 seconds.

Observations from the rate drop sensitivity are:

1. Maintaining a higher injection rate before shut-in results in higher water hammer peaks and troughs following shut-in (Joukowsky effect).
2. There are greater superpositioning effects on water hammer waveforms for the cases of relatively low injection rate before shut-in since the 1st rate drop is higher than the 2nd rate drop. For these cases, there is more energy from the 1st rate drop persisting through the 2nd rate drop. The rate drops are tabulated in Table 3. Rows with red font note the scenarios with observable superposition effects caused by the 1st rate drop.
3. The period is the same for all cases. This is expected as the well configuration is the same for all cases.

Table 3—Rate drops

Initial rate (bbl/min)	1 st rate drop (bbl/min)	2 nd rate drop, at shut in (bbl/min)
66	26	40
66	31	35
66	36	30
66	41	25
66	46	20
66	51	15

The recommendation is to have equivalent rate reductions or to have the last rate reductions to be higher than the prior rate reduction to minimize the superpositioning effect on the water hammer signature following shut-in.

Sensitivity Analysis #4.: The following 3 cases evaluate the effect of varying the number of equal-duration rate drops on the water hammer signature. The results are shown in Fig. 39.

1. Start at 100 bbl/min, four 25 bbl/min drops, each held for 30 seconds
2. Start at 100 bbl/min, three 25 bbl/min drops, last rate at 5 bbl/min, each held for 30 seconds
3. Start at 100 bbl/min, go half rate (50 bbl/min), hold for 30 seconds

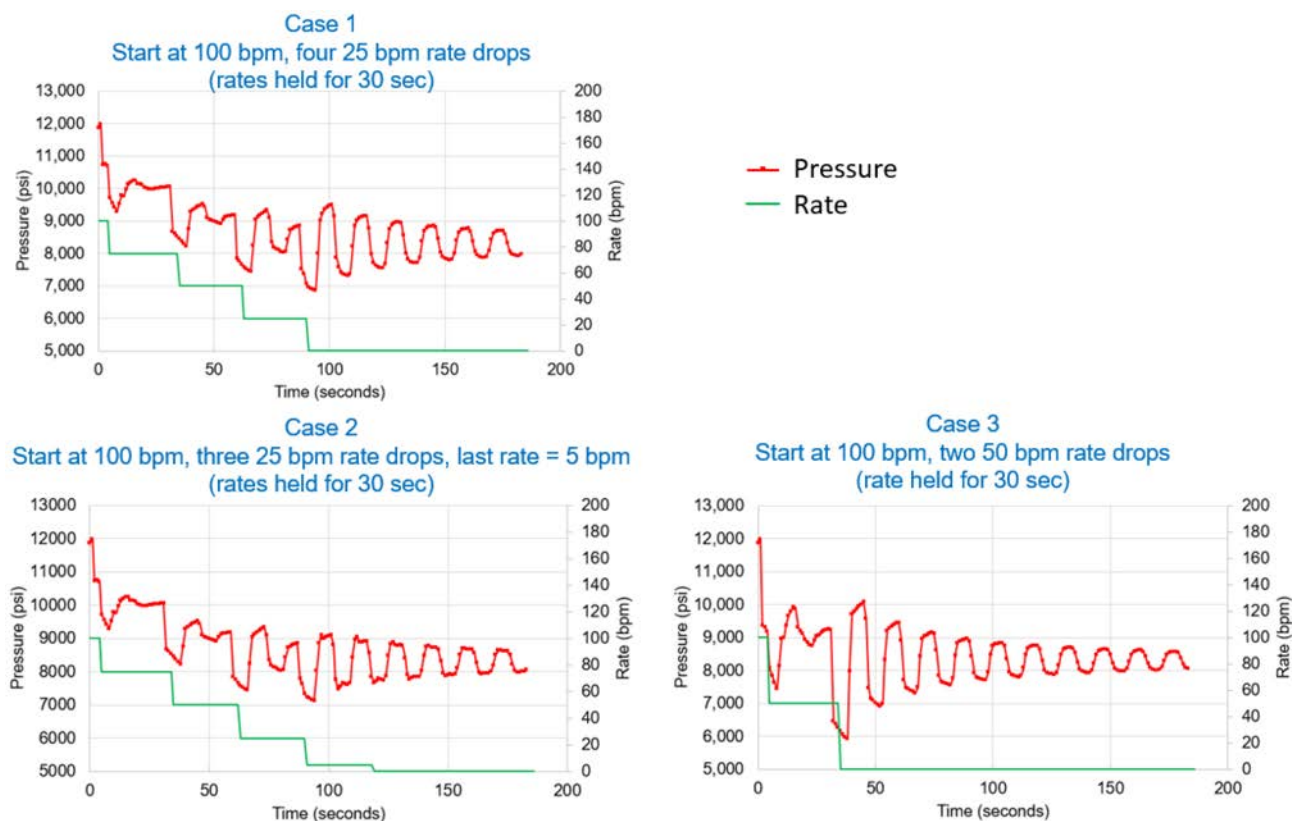


Figure 39—Sensitivity on the number of step-downs and step-down rate

Observations from this step-down sensitivity are:

1. Comparing Case 1 and 2, if a 5 bbl/min step-down is conducted after a larger step-down (20 bbl/min step-down), the prior water hammer signature covers the water hammer signature from the 5 bbl/min step-down. For the water hammer analysis of Case 2, the 5 bbl/min rate slightly reduces the peak/trough magnitude and water hammer shape compared to Case 1. For Case 2, the shut-in water hammer analysis could be considered to start after the 20 bbl/min step-down since the 5 bbl/min step-down had minimal effect on the water hammer signature.
2. For Case 1 and 2, a superpositioning effect is seen for each 25 bbl/min step-down.
3. Both 25 and 50 bbl/min rate reductions with 30 second duration provide clear water hammer signatures. A 25 bbl/min rate drop seems to be sufficient for analysis.

Recommendations:. The recommendations are:

1. The last rate step should be at least 25 bbl/min to generate a clear water hammer signature.
2. Avoid stepping down the rate to 5 bbl/min.
3. If the service company prefers to use multiple rate step-downs to lessen the impact of shut-down on the pumping equipment, the last step-down should be at a rate of 25 bbl/min or greater.
4. The duration of rate steps should be a minimum of 30 seconds to minimize superpositioning effects of multiple water hammer pulses.
5. **KEY POINT** – Performing the step-down in a consistent way is the most beneficial measure for obtaining meaningful comparisons of water hammer signatures across multiple treatment stages.

Perforation Friction

Three cases were simulated for the perforation friction sensitivity analysis (450, 1000, 1500 psi perforation friction). The injection rate starts at 66 bbl/min, is dropped to 33 bbl/min and held for 30 seconds, and then dropped to zero rate to initiate the shut-in period. The results of the evaluation are shown in Fig. 40.

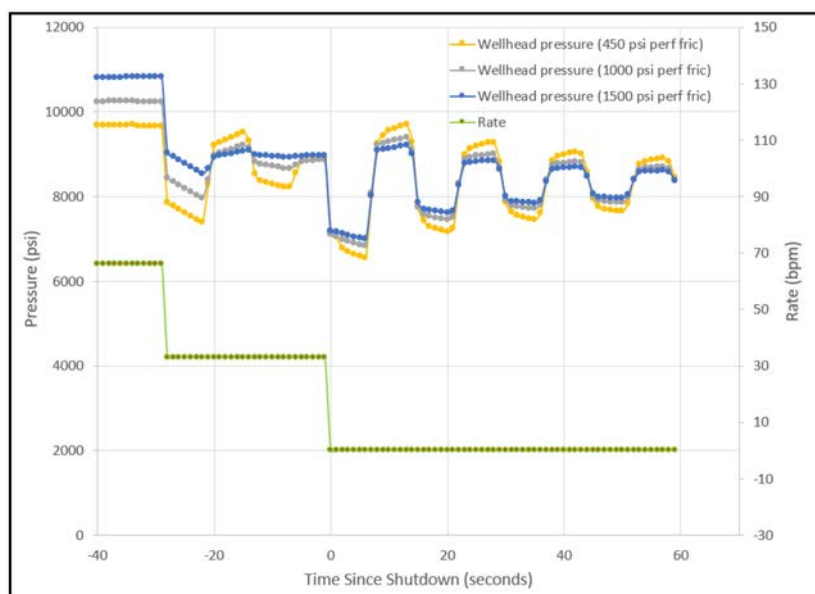


Figure 40—Perforation friction sensitivity analysis

Observations from the perforation friction sensitivity are:

1. As expected, the highest perforation friction case (1,500 psi) had the highest wellhead pressure while pumping at full injection rate.

2. Lower perforation friction equates to higher peaks and deeper troughs as compared to higher perforation friction.
3. During the 30-second hold period of the rate step-down (33 bbl/min), higher perforation friction correlates with greater dampening of the water hammer signature.
4. During the shut-in period, the 1,000 and 1,500 psi perforation friction cases exhibited minimal to no superposition effect from the water hammer signature created from the initial rate reduction. This outcome was the result of signal dampening. For the 450-psi case, there is a slight superposition effect as the water hammer signature from the initial rate reduction was not completely dampened.
5. Initially, the difference between the water hammer peaks and troughs were:
 - i. 400-500 psi between the 450 and 1,500 psi perforation friction cases.
 - ii. 200 psi between the 1,000 and 1,500 psi perforation friction cases.
6. The pressure difference among the three cases decreases over time with the decay of the water hammer signature.
7. Additional perforation friction dampens the water hammer signature only slightly, and not significantly.

Application of Water Hammer Analysis

Identification of Boundary Conditions

A case demonstrating differing boundary conditions is presented for two different types of operations in the same well. Fig. 41 shows the water hammer signature after a perforating event while Fig. 42 shows the water hammer signature after the main hydraulic fracturing treatment. The perforation depth was 9,416 ft.

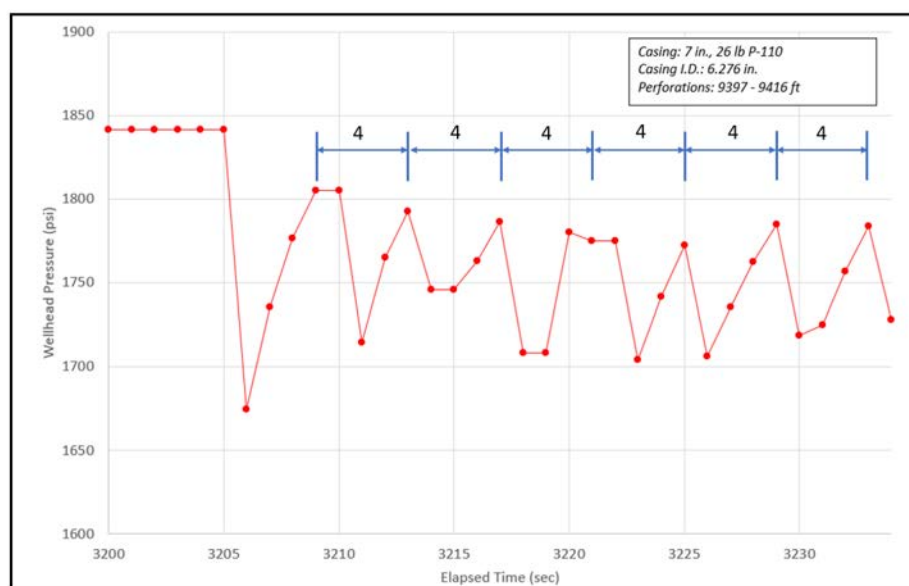


Figure 41—Water hammer after perforation detonation event

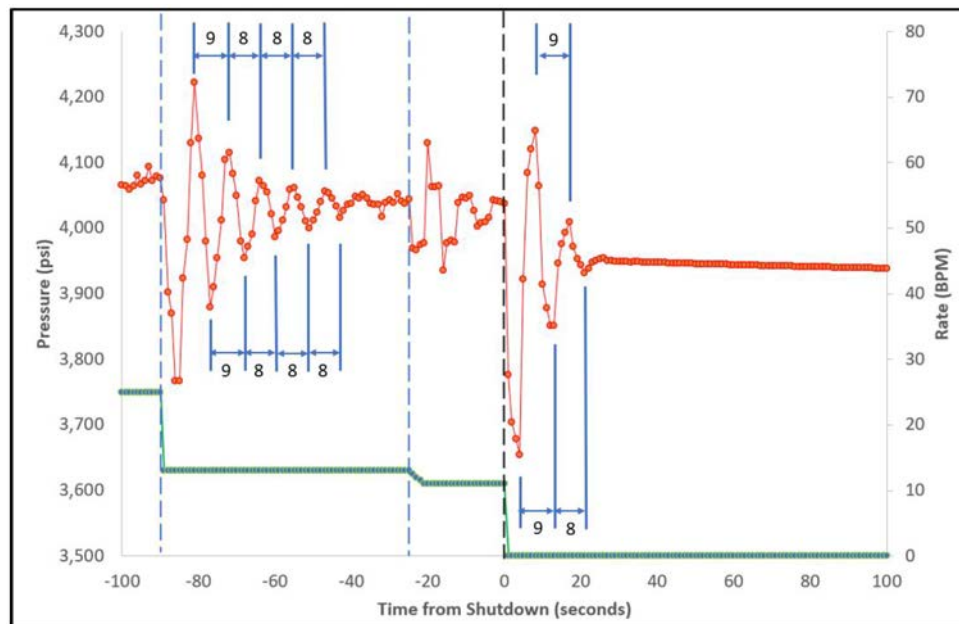


Figure 42—Water hammer after hydraulic fracturing treatment

For the perforation event, the period was 4 seconds. Assuming a fluid speed of sound of 5,000 ft/s, the boundary condition factor is 2.1 seconds or approximately 2 as per Equation 3 ($4 \text{ sec} \times 5,000 \text{ ft/s} / 9,416 \text{ ft} = 2.1$). A boundary condition factor of 2 denotes that the boundary condition is a closed inlet and closed outlet. There was nil fracture capacity at the perforations.

For shut-in period following the fracturing treatment performed on this well, the period was 8 to 9 seconds. This is double the period for the perforating event. The boundary condition factor was about 4, indicative of a closed inlet and constant pressure outlet. This denotes that the well was in communication with a large capacity hydraulic fracture system. A question still to be further understood is how much fracture capacity is required to switch from a boundary condition factor of 2 to 4.

Another case demonstrating differing boundary conditions is presented for two treatment stages in the same well. Stage 6 had a successfully completed hydraulic fracture treatment with a period of 15 seconds, as shown in Fig. 43. The boundary condition factor for this treatment was 4. Stage 7 had a screen out which resulted in a period of 7 to 8 seconds, as shown in Fig. 44. This was half the period of Stage 6, indicative of a boundary condition of a closed inlet and a closed outlet. The screen out in Stage 7 completely bridged the wellbore and fracture system near the perforations, disconnecting the travel path of the water hammer from the high capacity hydraulic fracture system.

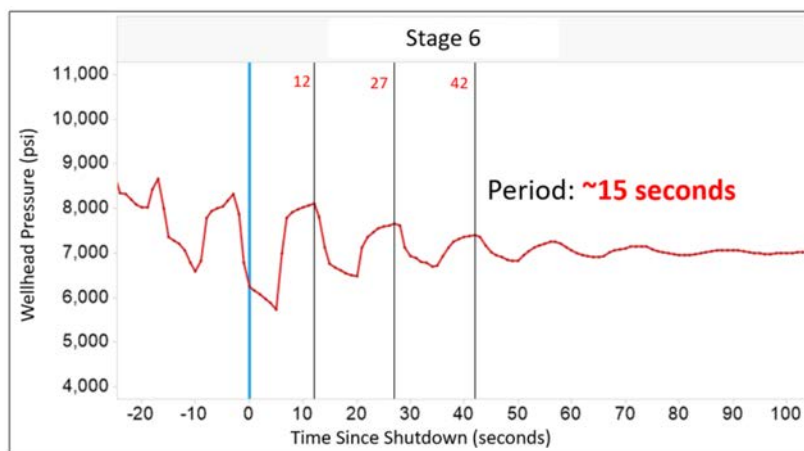


Figure 43—The corresponding water hammer of a treatment with no operational issues (Stage 6).

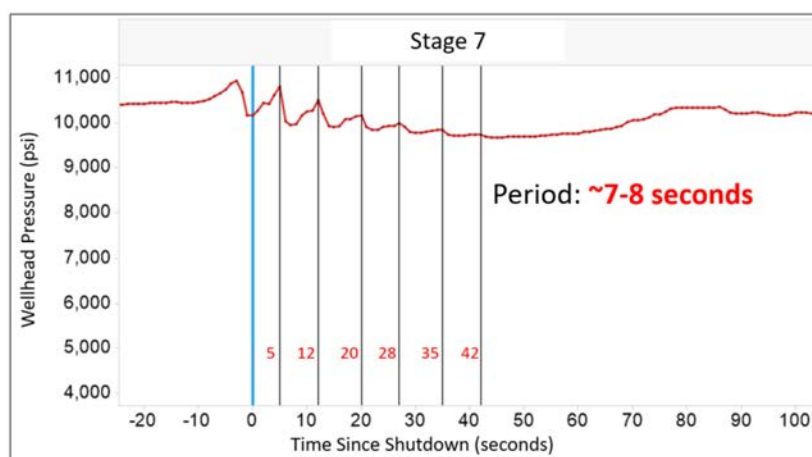


Figure 44—The corresponding water hammer of a treatment with a screen out event (Stage 7).

Identification of Treatment Stage Isolation

Water hammer boundary condition calculations can provide indicators for evaluating isolation among treatment stages in pumpdown diagnostic testing. As described in SPE-201376 (Cramer et al. 2020), pumpdown diagnostics are performed during plug-and-perf horizontal well treatments when isolating a previous treatment stage and perforating a new interval, and they consist of the following activities.

1. Pump down the frac plug and perforating guns.
2. Pressure test the frac plug.
3. Perforate the first cluster, closest to the toe end of the well.
4. Conduct an injectivity test.
5. Perforate the remaining clusters.

For Fig. 45 - 47, the activity numbering is identified on the charts with associated color coding.

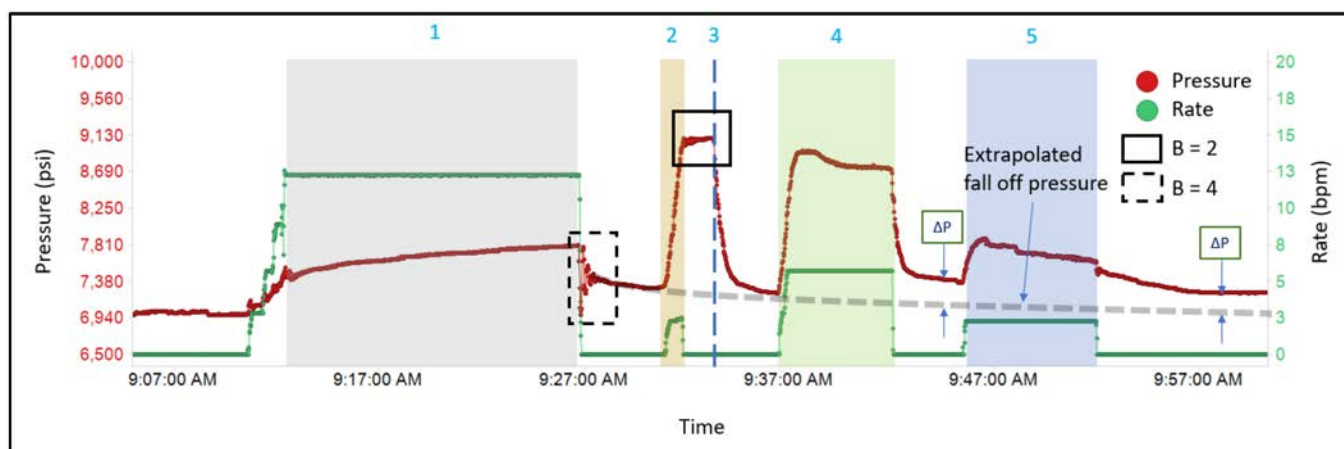


Figure 45—Pumpdown diagnostics showing stage isolation.

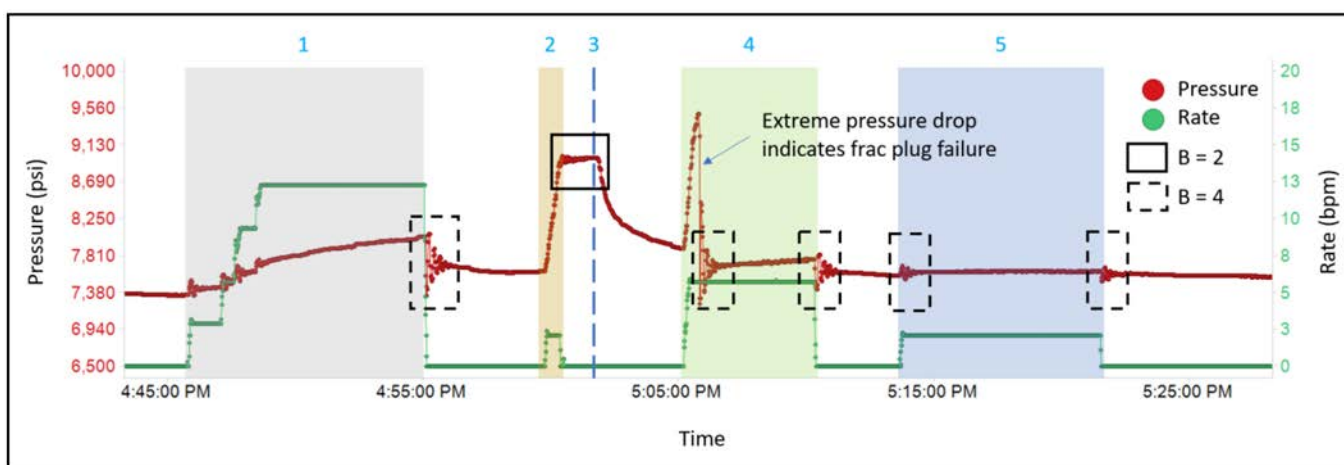


Figure 46—Pumpdown diagnostic testing showing frac plug failure (loss of stage isolation).

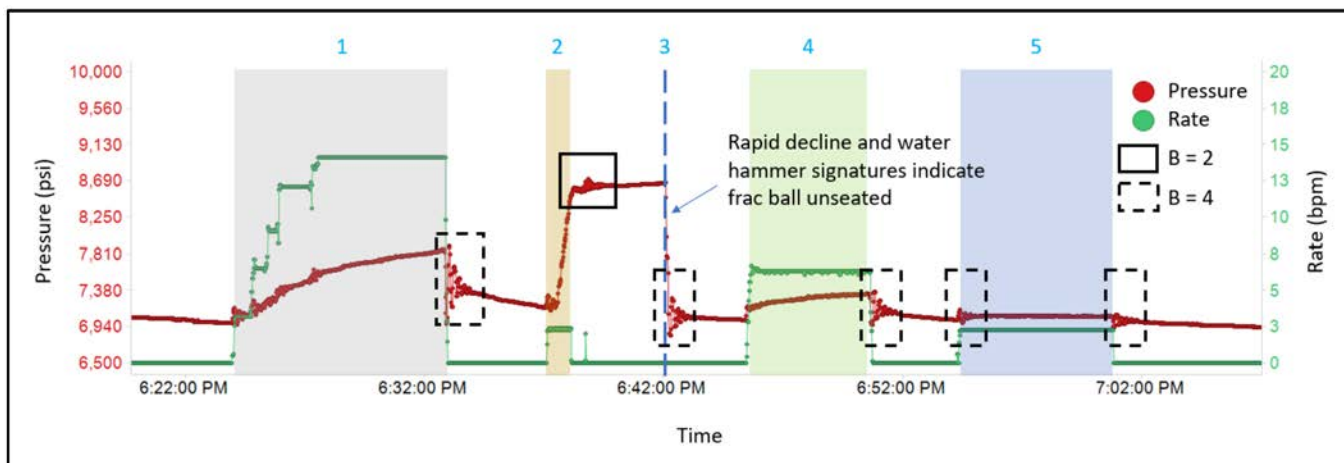


Figure 47—Pumpdown diagnostics showing an unseated frac ball (loss of stage isolation).

Fig. 45 - 47 are pumpdown diagnostic plots for three stages in the same well. Fig. 45 shows a case in which testing confirmed the newly-perforated stage was isolated from the prior treatment stage. During testing, two water hammer signatures occurred, one after the pump down injection and the other after the frac plug pressure test.

- The dashed line box around the first water hammer signature after the pump down denotes that the boundary condition factor was 4 (closed inlet and constant pressure outlet). This notes there was a connection to a large fracture capacity (the previous stage that was hydraulically fractured).
- The solid line box around the second water hammer signature after the frac plug pressure test denotes that the boundary condition factor was 2 (closed inlet and closed outlet). This confirms that at this point the wellbore was a closed system with no leakage past the frac plug. The ball successfully seated in the frac plug and a water hammer pulse was generated from the sudden rate termination.

After the last three activities (perforation of the first cluster, injectivity test, and perforation of the remaining clusters), there was a gradual decline in pressure with no water hammer signature. Additionally, the fall-off pressures are greater than the extrapolated pumpdown pressure fall-off trendline, when there was still connectivity to the prior stage. When combined, these indications strongly confirmed that the new treatment interval was isolated from the previous treatment interval.

Fig. 46 shows a frac plug failure occurring during pumpdown operations. This is indicated by the extreme pressure drop at the start of the injectivity test. For this stage, there were six water hammer signatures. The first water hammer signature after the pump down had a boundary condition factor of 4, indicating wellbore connection to the large fracture capacity of the prior stage. The second water hammer signature after the frac plug test had a boundary condition factor of 2, indicating that the frac plug achieved isolation from the prior stage. The last four water hammer signatures that occurred after the frac plug failure all had a boundary condition factor of 4, confirming loss of isolation and connection once again to the large fracture capacity of the prior stage.

Fig. 47 shows a stage where a frac ball unseated as indicated by a rapid pressure decline after the perforation of the first cluster. The water hammer signatures for this scenario were the same as the frac plug failure scenario, showing that stage isolation was lost.

Identification of Casing Failure Depth

For the following case, treatment stage 1 of a well was performed with no noticeable issues. The average injection rate and surface treating pressure for this stage were 65 bbl/min and 9,000 psi, respectively. Treatment stage 2 initially exhibited similar rate and pressure behavior as stage 1. However, 25 minutes into the treatment, the rate and pressure changed significantly, as the rate increased to 90 bbl/min and the surface treating pressure decreased to 7,500 psi. This change indicated that the depth of the fluid moving out of the wellbore could be significantly lower than expected, potentially as a result of a casing failure located far from the perforated interval.

Fig. 48 compares the water hammer signature from stage 1 and 2. The period for stage 1 was 20 seconds; the period for stage 2 was 9 seconds.

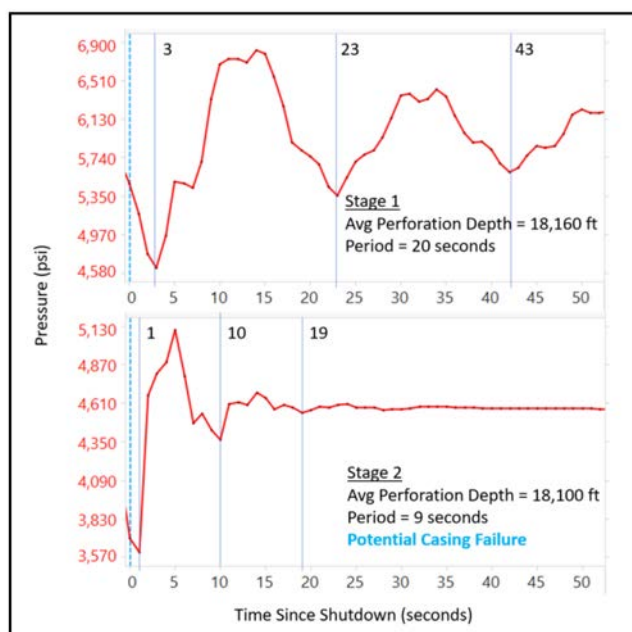


Figure 48—Comparison of stage 1 and stage 2 water hammer period

The boundary condition for this case is closed inlet and constant pressure outlet, so the boundary condition factor was 4. Assuming the fluid speed of sound was 5,000 ft/s, the following measured depths were calculated for periods of 8, 9, and 10 seconds (period sensitivity of ± 1 second to account for the data collection frequency of 1 second). Measured depth of the flow exit is calculated by multiplying the period by the fluid speed of sound and then dividing by the boundary condition factor and the results are shown in Table 4.

Table 4—Measured depth of flow exit (potential casing failure depth)

Period (seconds)	8	9	10
Measured depth of flow exit (ft)	10,000	11,250	12,500

Later, caliper logs and downhole camera imaging identified casing pits at depths ranging from 10,125 to 10,153 ft, intermediate to the calculated depths associated with 8 and 9 seconds of period.

Determination of Excess Period (Excess Length)

In Fig. 48, the period predicted for the perforation depth (18,160 ft) and 5,000 ft/s fluid speed of sound was 14.5 seconds ($18,160 \text{ ft} / 5,000 \text{ ft/s} \times 4$). However, the water hammer signature from stage 1 showed a period of 20 seconds. The excess period was 5.5 seconds ($20 \text{ s} - 14.5 \text{ s}$). Excess period can also be expressed as excess length. The predicted length for 20 second period is 25,000 ft ($20 / 4 \times 5,000$). Correspondingly, the excess length is 6,840 ft ($25,000 \text{ ft} - 18,160 \text{ ft}$). This is an increase of 38% in respect to period or length.

Further investigation is required to determine if excess period and the associated excess length value provide indications of hydraulic fracture dimensions or rather fluid/casing property anomalies that reduce the fluid speed of sound. Within the water hammer model described previously, additional length can be added to the perforation depth to account for the excess period. However, as compared to water hammer wave travel in casing, the speed of sound in hydraulic fractures is much slower, highly variable, and difficult to determine (Paige et al. 1992). Consequently, the above calculation for excess length should not be considered as equivalent to fracture length.

Case Study – Water Hammer Analysis in an Unconventional Reservoir

Using the methods described in the sections above, water hammer data was analyzed for 8,831 fracturing stages in 395 wells in a North America unconventional reservoir. The analysis focused on the relationship of water hammer characteristics with the completion design and resulting well productivity. Water hammer data was not available on all stages of every well due to data quality issues. For production analysis, only the wells with water hammer data available on at least 50% of the stages were evaluated.

A high-level summary of the findings from the analysis indicated the following.

- The water hammer decay rate is most affected by near-wellbore fracture surface area.
- A higher water hammer decay rate equates to contacting more near-wellbore fracture surface area.
- A very low water hammer decay rate correlates with lower well productivity.
- Low water hammer decay rates also correlate with long distance fracture-driven interactions (FDI), also known as frac hits.
- The water hammer decay rate becomes more variable as the total treatment volume for a well increases.

This study was limited to wells within a single field and geologic basin. The relationship of water hammer characteristics such as decay rate with well productivity observed in this field may not be the same in other geologic settings with differing rock properties or in-situ stress distributions.

For this analysis, the total number of water hammer periods was used as a proxy for the water hammer decay rate due to ease of calculation and its sufficiency for performing a straightforward comparison among fracturing stages. The terminology of water hammer oscillation characteristics is covered in Fig. 23. As indicated there, the decay rate is inversely proportional to the number of water hammer periods.

Interpretation of frac geometry and friction in the fracture

Impact of Friction on Water Hammer Decay Rate. As the water hammer pulse travels back and forth within the wellbore and hydraulic fracture system, friction causes it to dampen over time. There are three potential sources of friction that dampen water hammer pulses:

- Fluid viscosity
- Contact with surface area inside the wellbore
- Contact with surface area outside the wellbore

Of the three sources, friction due to contact with surface area outside the wellbore and primarily within the hydraulic fracture system is typically dominant and is the primary reason for water hammer decay rates varying for fracturing stages having the same treatment design.

High viscosity fluids, such as crosslinked gel, will cause a water hammer signature to dampen faster. For analysis purposes, this is not typically an issue because in any given well, the same fluid type is used for each fracturing stage. However, this needs to be accounted for when comparing water hammer data between wells that were treated with different fluid types. Even though crosslinked gel stages are flushed with slick water, when the water hammer pulse exits the wellbore, it will travel through the crosslinked gel filling the fractures which can influence the water hammer decay rate.

In this dataset, 5,484 stages were completed with crosslinked gel and 3,347 stages were completed with slick water. When comparing stages that had the same treatment size (2,600 lbs of proppant/ft), it was found that the difference in number of water hammer periods between crosslinked gel and slick water is roughly

0.5 periods, as shown in Fig. 49. When compared to the range of values for number of water hammer periods, a difference of 0.5 periods is small but not negligible.

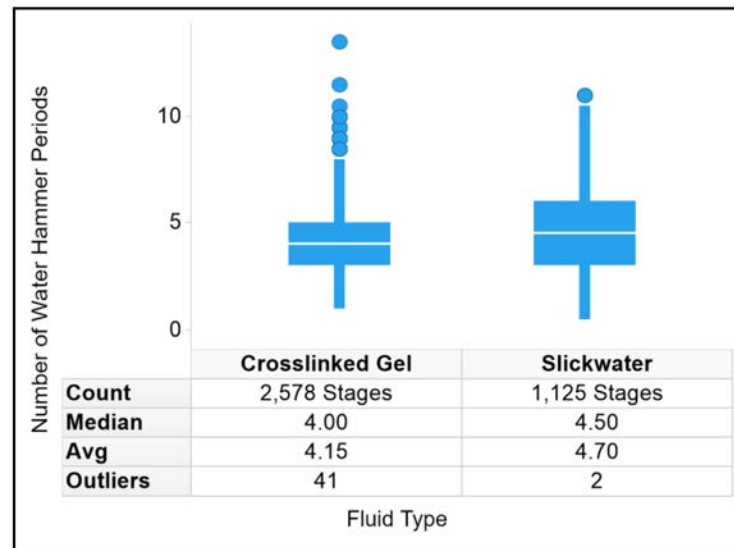


Figure 49—Completions fluid type versus number of water hammer periods

When evaluating water hammer data for all stages, there is not a clear trend between number of water hammer periods and stage depth. The primary reason for this is that friction within the hydraulic fracture system can have the dominant effect on friction and thus the water hammer decay rate during the post-treatment shut-in period. This is primarily the result of differences in surface area as demonstrated in the following hypothetical example.

A wellbore consisting of 5-1/2 in. casing at a measured depth of 20,000 ft has an internal surface area of 24,450 ft². The cumulative fracture surface area for a stage with 10 perforation clusters, each connected to one smooth-walled, planar fracture extending 75 ft radially from the wellbore is 176,700 ft². In this example, the fracture surface area is more than seven times greater than the wellbore surface area. It is a conservative estimate of the potential difference. Hydraulic fractures typically extend much farther than 75 ft radially from the wellbore. Field studies indicate that hydraulic fracture systems can be complex, with much greater surface area and fracture-width variation than the simple case presented above (Rateman et al. 2019).

The above exercise is continued to demonstrate the relative effects of variations in wellbore and fracture system components on surface area and thus friction. The difference in surface area between the two-fold difference in measured depth of 10,000 and 20,000 ft is 12,225 ft². The difference in surface area between a stage that treated half fracture per cluster with a stage that treated one fracture per cluster (two-fold difference in the number of fractures) is a conservatively estimated difference of 88,350 ft². Variation in fracture system properties will have a greater impact on surface area and consequently on friction and water hammer decay rate.

There are qualifications to the above assessment. The data used for this analysis was primarily on wells with 5-1/2 in. 23 lb/ft casing using plug-and-perf completions, with measured depths varying between 11,000 ft and 21,000 ft among all fracturing stages. Perforation friction typically has a minor influence on water hammer characteristics in this style of completion. Yet the situation may be somewhat different for other completion types. For instance, as reported by Iriarte et al. (2017), treatments using the ball-actuated sliding sleeve method of treatment sequencing exhibit relatively high water hammer decay rates due to the sleeve ball seats acting as baffles as the water hammer pulses flow in and out of the wellbore.

Water Hammer Decay Rate as an Indication of Near-Wellbore Fracture Geometry. As postulated by Ciezobka et al. (2016), water hammer dampening or decay is affected by the degree of fracture connectivity with the wellbore. Being in contact with a greater number of fractures results in more rapid signal dampening or decay since friction is proportional to fracture surface area and complexity. This case is exemplified in comparing Fig. 50a. and Fig. 50b. with Fig. 50c. and Fig. 50d. Being in contact with fewer fractures or a less complex fracture network results in less friction and slower signal dampening or decay.

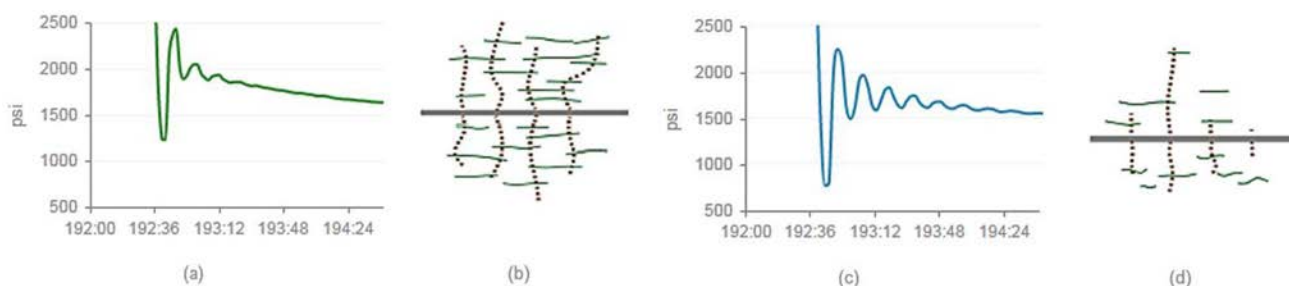


Figure 50—Proposed relationship of water hammer decay rate with contacted fracture area (Iriarte et al. 2017)

Two relationships observed in analyzing the case study data support the above postulation. Wells with very low water hammer decay rates typically have poorer well productivity. Low water hammer decay rates also correlate with instances of long-distance FDI's, i.e., frac hits resulting from creating fewer and longer hydraulic fractures.

A related observation was that the water hammer decay rate became more variable as the volume of fracturing fluid and proppant per lateral foot treatment size increased. This relationship is shown in Fig. 51. Increases in treatment volume were the result of increasing the number of perforation clusters (fracture initiation points) per foot of lateral or increasing the treatment volume (proppant and fluid) per cluster. Although these tactics can lead to increased fracture density, cumulative fracture surface area and water hammer decay rate, they can also lead to increased communication among clusters and proppant bridging within less advantaged fractures (Cramer et al. 2020). The latter outcomes will reduce the number of active hydraulic fractures, forcing more volume into fewer fractures which decreases cumulative fracture surface area, friction acting on the water hammer pulse, and the water hammer decay rate.

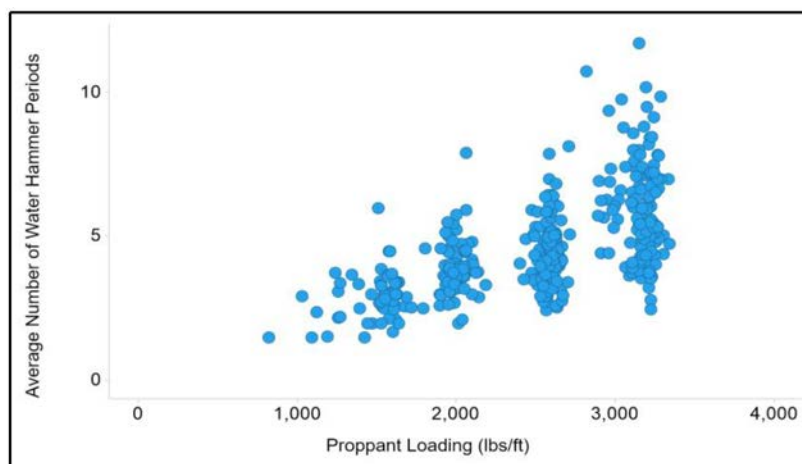


Figure 51—Average number of water hammer periods per well versus proppant volume.

Of all variables analyzed, treatment volume per foot of lateral had the strongest correlation with the number of water hammer periods per stage. As shown in Fig. 52, wells characterized by low average

water hammer decay rates (red bar) typically had much longer-reaching FDI's. FDI's were determined by identifying pressure increases in passive offset wells that were synchronous with treatments being performed in the active analyzed well. The data in Fig. 52 is from 68 wells that had the same perforation cluster spacing, number of clusters and proppant volume for each treatment stage. The cutoff used for low decay rate was six or more water hammer periods per treatment stage and the cutoff for high decay rate was four or fewer periods per treatment stage. This data suggests that for a given treatment volume, treatments with low water hammer decay rates are associated with the creation of fewer, longer, less complex fractures, resulting in less cumulative fracture surface area.

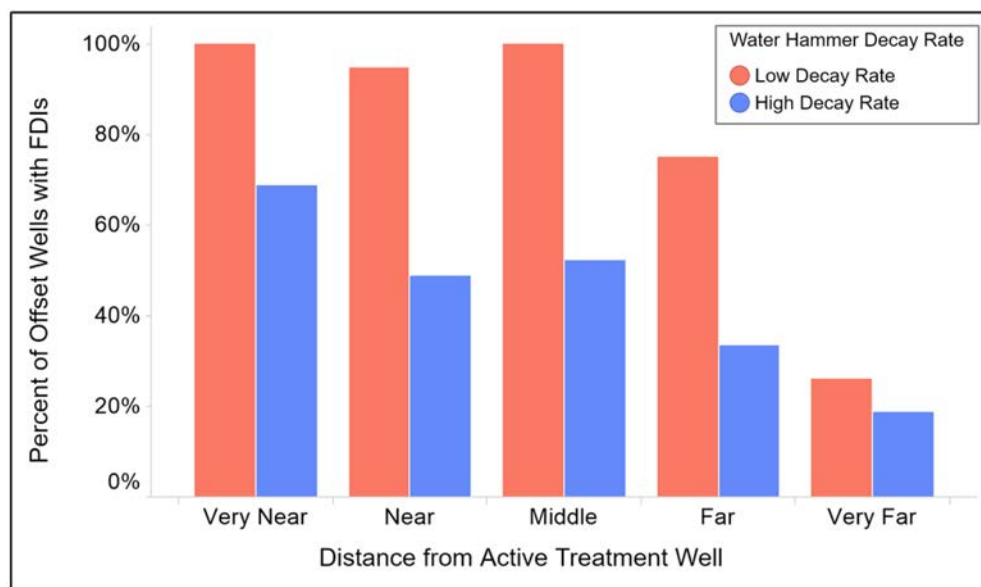


Figure 52—FDI's versus distance from the well being actively treated.

Relationship Between Water Hammer and Well Productivity

Wells that have very low water hammer decay rates commonly exhibit lower well productivity, underperforming by 10% to 20% as compared to wells with higher water hammer decay rates. The cutoff used for determining a very low decay rate depended on the size of the treatment. For the wells in this data set, treatments characterized by 3,200 lbs of proppant/ft of lateral were considered to have a very low decay rate if it had an average of seven or more water hammer periods per treatment stage. Treatments characterized by 2,600 lbs of proppant/ft of lateral, six or more water hammer periods per treatment stage was classified as a very low decay rate.

Fig. 53 and Fig. 54 show the relationship of well performance to water hammer decay for the two treatment-volume categories. Type curve expectation is the 35-year estimated ultimate recovery (EUR) for each well. It is based on a correlation of geologic, petrophysical and treatment characteristics with historical well productivity in the area. The results for both groupings show that well productivity is lower on wells that have longer-lasting water hammers, and substantially lower for instances of very low water hammer decay rate as defined previously.

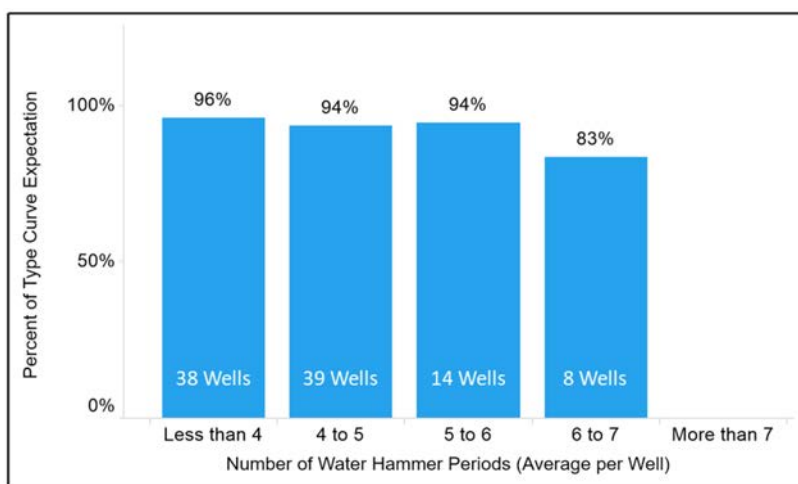


Figure 53—Well performance versus average number of water hammer periods for wells with 2600 lbs/ft proppant.

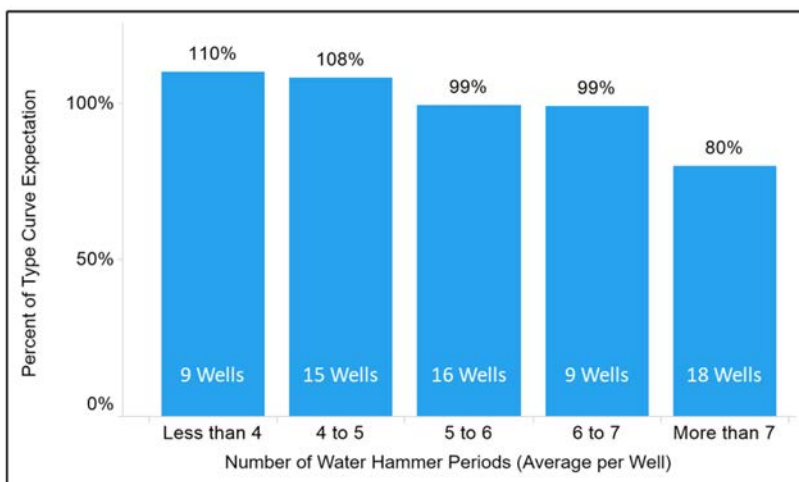


Figure 54—Well performance versus average number of water hammer periods for wells with 3200 lbs/ft proppant.

Conclusions

- Setting data requirements with service companies and data aggregation companies will lead to obtaining high quality data for water hammer analysis.
- The numerical water hammer model presented in the paper provides insight into physical processes associated with water hammer waveforms and is a vehicle for sensitivity testing of wellbore and treatment variables to evaluate the corresponding effect on water hammer signatures.
- Using a consistent injection-rate step-down process at the end of fracturing treatments leads to more reliable results when comparing water hammer characteristics among multiple treatments and wells.
- The water hammer decay rate is affected by pipe friction and friction in hydraulic fracture network.
 - Continuing to pump during a water hammer, as is done during the injection rate step-down process at the end of treatments, increases the decay rate.
 - During the shut-in period, there is no active pumping. However, there is still friction from the back-and-forth movement of fluid within the wellbore/fracture system that affects decay rate.

- When fluid viscosity, friction reducer effectiveness, and pipe geometry are consistent among treatments being evaluated, pipe friction has a smaller impact on variations in the water hammer decay rate as compared to friction in the fracture network.
- The water hammer decay rate appears to be mostly influenced by the fracture surface area near the wellbore. High decay rates are an indication of a large amount of near-wellbore fracture surface area and low decay rates indicate less near-wellbore fracture surface area.
- For the wells analyzed in the unconventional reservoir case study data set, low water hammer decay rates correlated with relatively lower well productivity and long FDI's.
- Optimal water hammer characteristics as related to well productivity may vary across fields and completion design types. Consequently, water hammer comparative analysis studies should be limited to specific completion styles, and geographic and geologic settings.

Nomenclature

A_k	cross-sectional area in momentum grid at position k
y_k	elevation in momentum grid at position k
z_k	distance in momentum grid at position k
$\bar{z}_k$	distance in mass grid at position k
ρ_k	density in momentum grid at position k
$\bar{\rho}_k$	density in mass grid at position k
P_k	pressure at position k
Δz_k	length of momentum grid at position k
$\Delta \bar{z}_k$	length of in mass grid at position k
$a\Delta \bar{z}_k$	volume of in mass grid at position k
$\dot{m}_k$	mass rate at position k
bpm	barrel per minute; bbl/min
CSV	comma separate values
dP	change in pressure
d	derivative
FDI	fracture driven interactions
ISIP	instantaneous shut-in pressure
t	time
TVD	true vertical depth

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